Computer Networks

Introduction

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Data Communication

The term telecommunication means communication at a distance. The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data. Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Topics discussed in this section:

Components
Data Flow

Reference: Behrouz A. Forouzan, *Data Communications and Networking*, Article 1.1

Components

1. Message
2. Sender
3. Receiver
4. Transmission Medium
5. Protocol

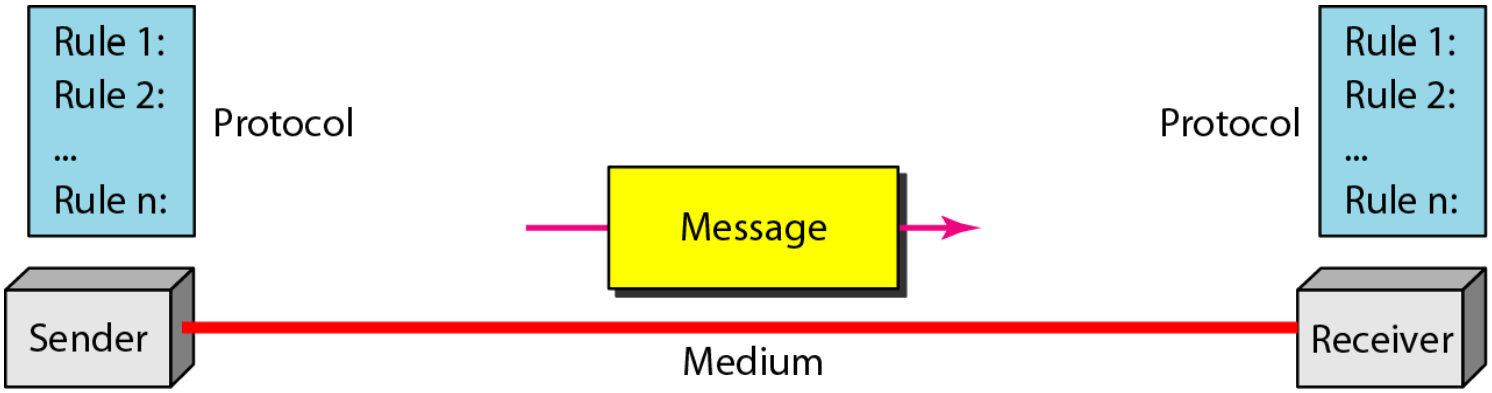


Figure 1.1 *Five components of data communication*

Data Flow

Communication between two devices can be simplex, half-duplex, or full-duplex

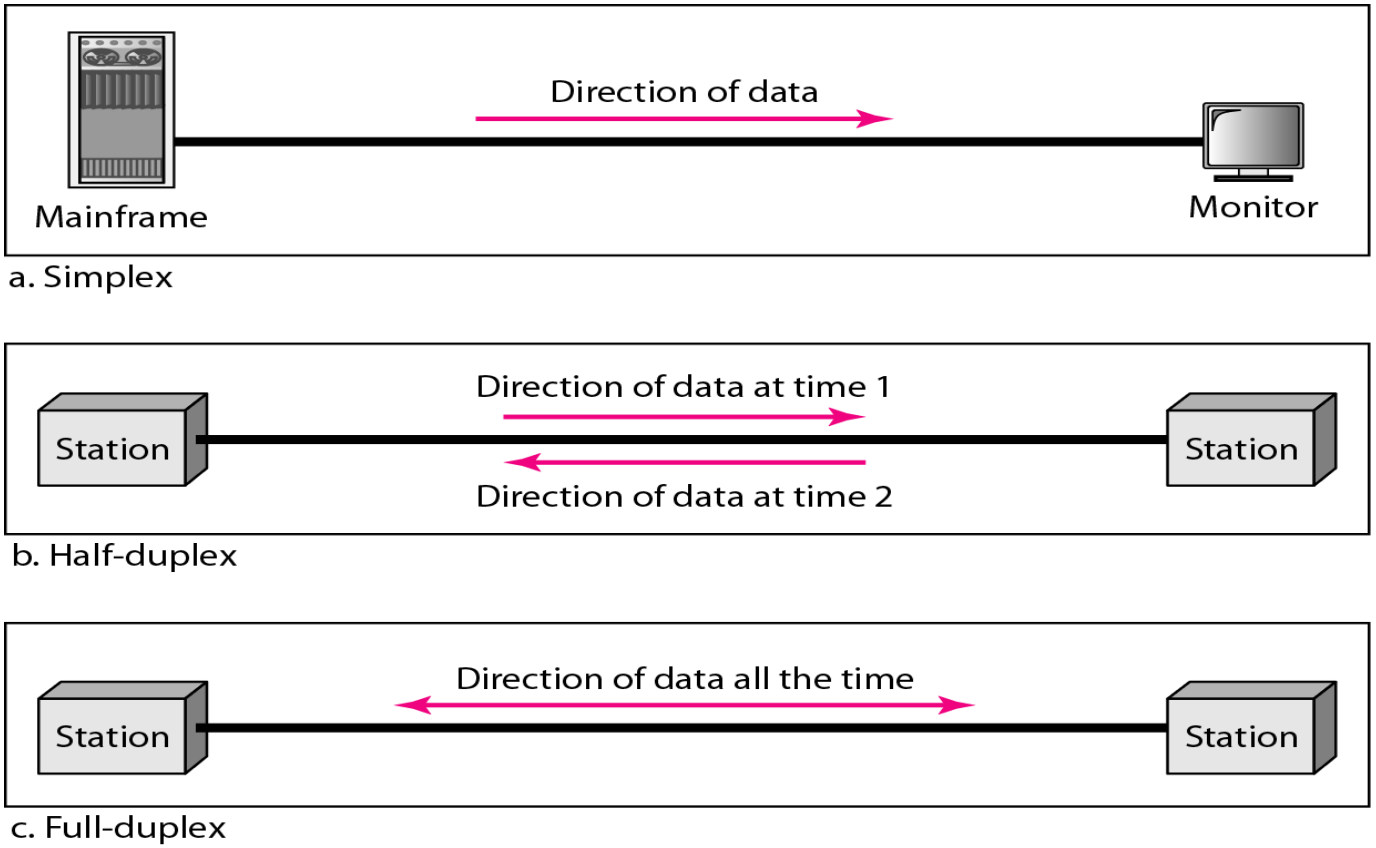


Figure 1.2 ***Data flow (simplex, half-duplex, and full-duplex)***

Networks

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

A computer network is the connection between more than one computer.

Reasons of using computer networks:

- Data Sharing
- Resource sharing

Topics discussed in this section:

Distributed Processing
Network Criteria
Physical Structures
Categories of Networks
Interconnection of Networks: Internetwork

Reference: Behrouz A. Forouzan, *Data Communications and Networking*, Article 1.2

Distributed Processing

- Most networks use distributed processing, in which a task is divided among multiple computers.
- Instead of one single large machine being responsible for all aspects of a process, separate computers (usually a personal computer or workstation) handle a subset.

Network Criteria

A network must be able to meet a certain number of criteria.

Performance:

- Performance can be measured in many ways, including transit time and response time.
- Transit time is the amount of time required for a message to travel from one device to another. Response time is the elapsed time between an inquiry and a response.
- Performance is often evaluated by two networking metrics: throughput and delay. We often need more throughput and less delay.

Reliability:

- Network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.

Security:

- Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses

Physical Structures

- Type of Connections
- Physical Topology

Type of Connections

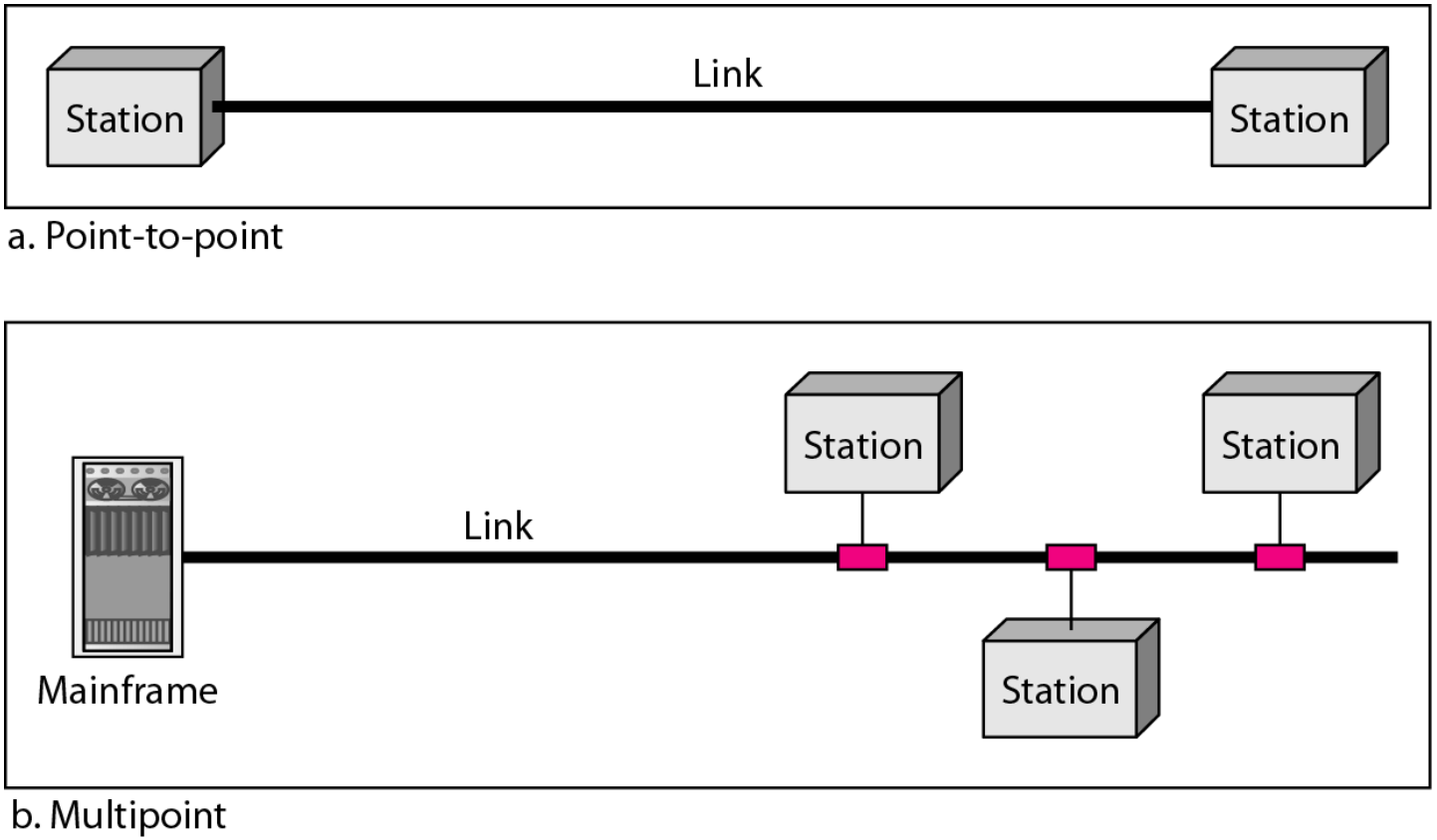


Figure 1.3 *Types of connections: point-to-point and multipoint*

Physical Topology

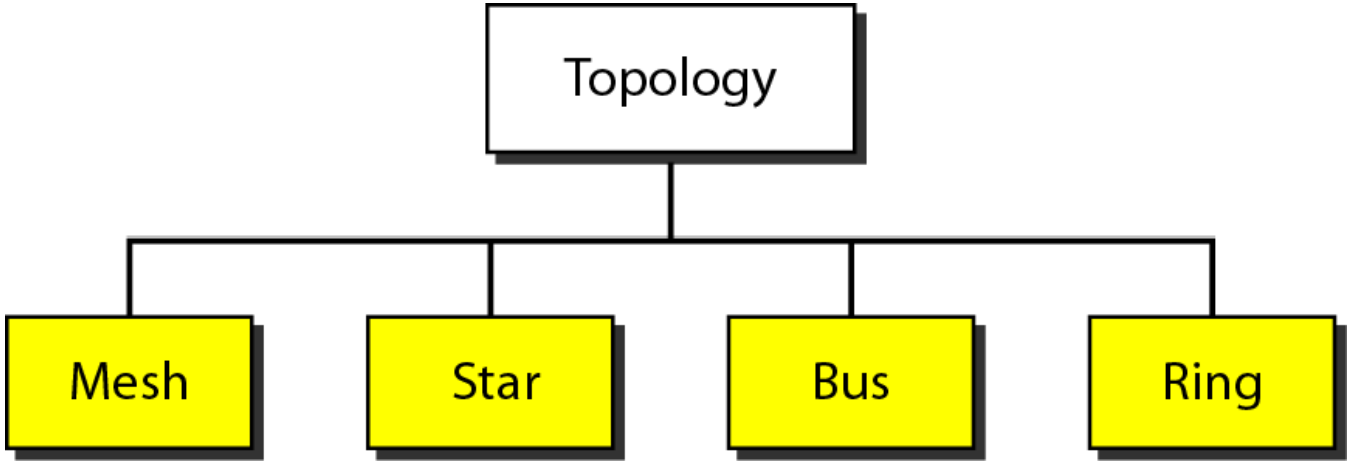


Figure 1.4 *Categories of topology*

Physical Topology

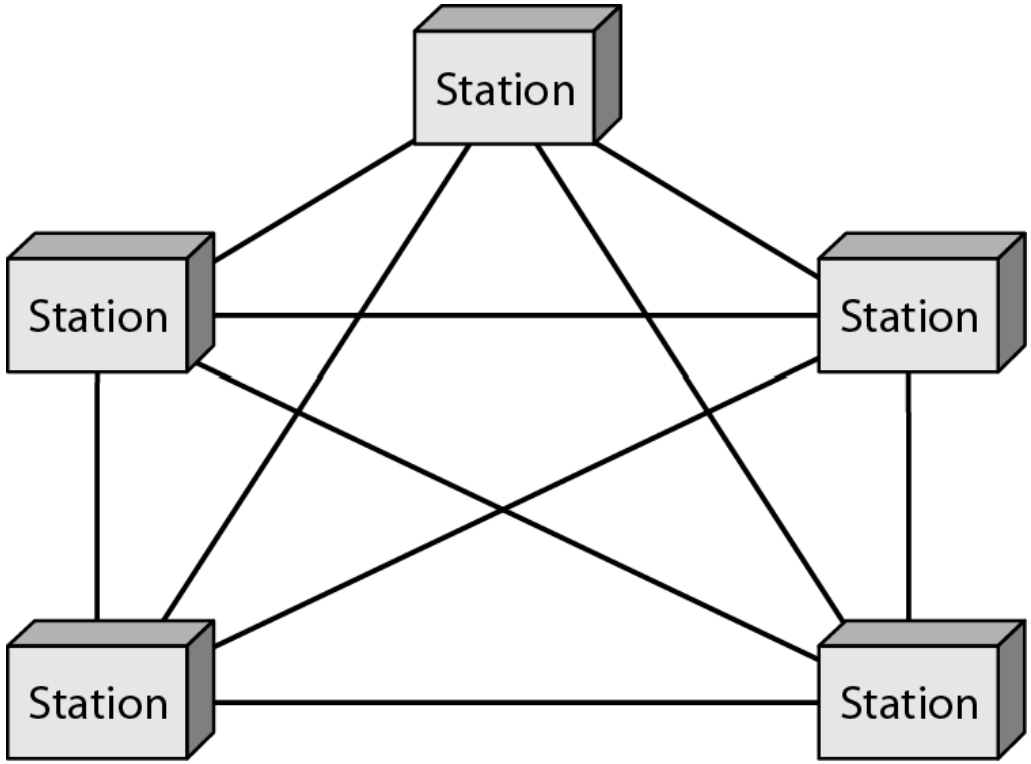


Figure 1.5 *A fully connected mesh topology (five devices)*

Physical Topology

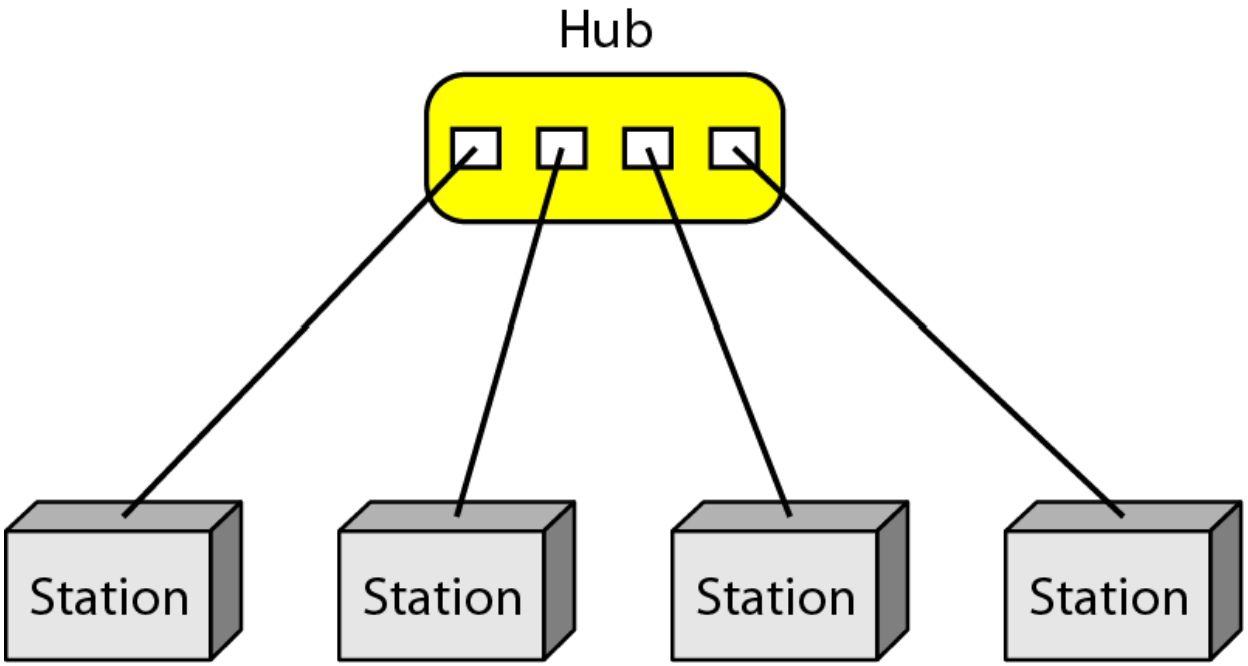


Figure 1.6 *A star topology connecting four stations*

Physical Topology

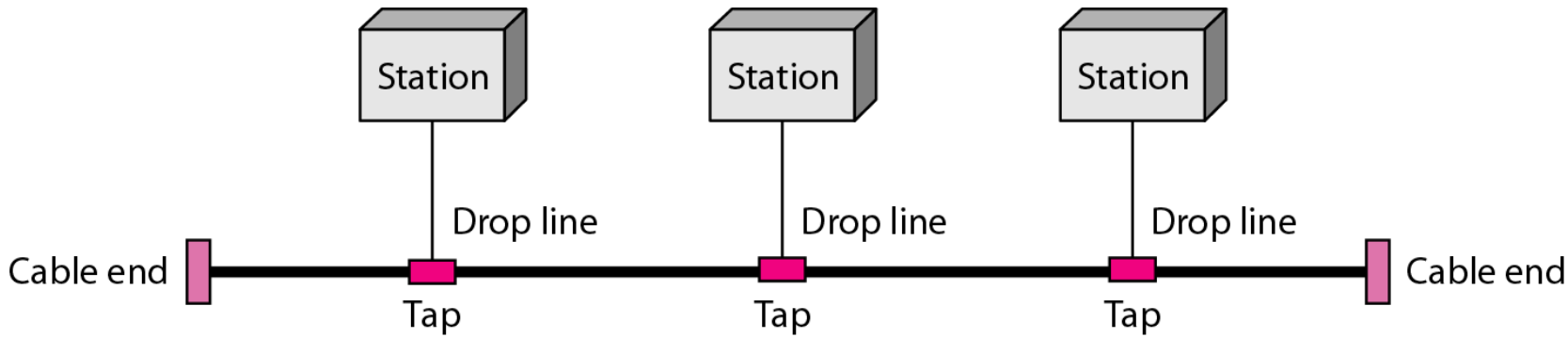
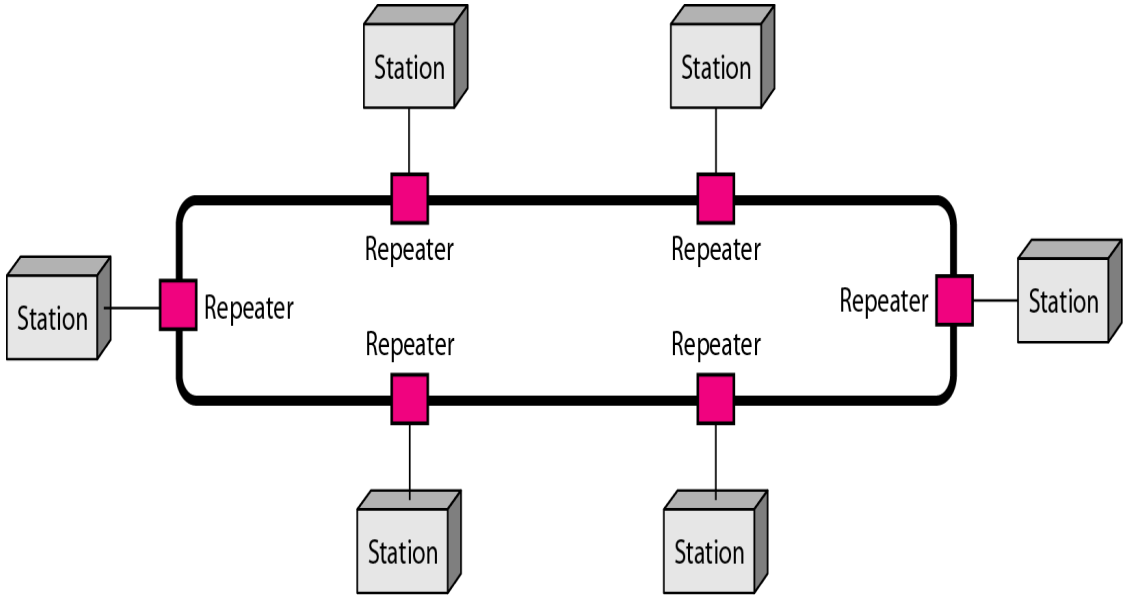


Figure 1.7 ***A bus topology connecting three stations***

Physical Topology



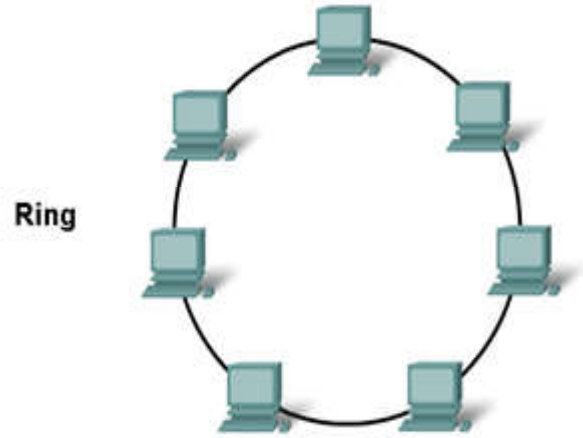


Figure 1.8 *A ring topology connecting six stations*

Physical Topology

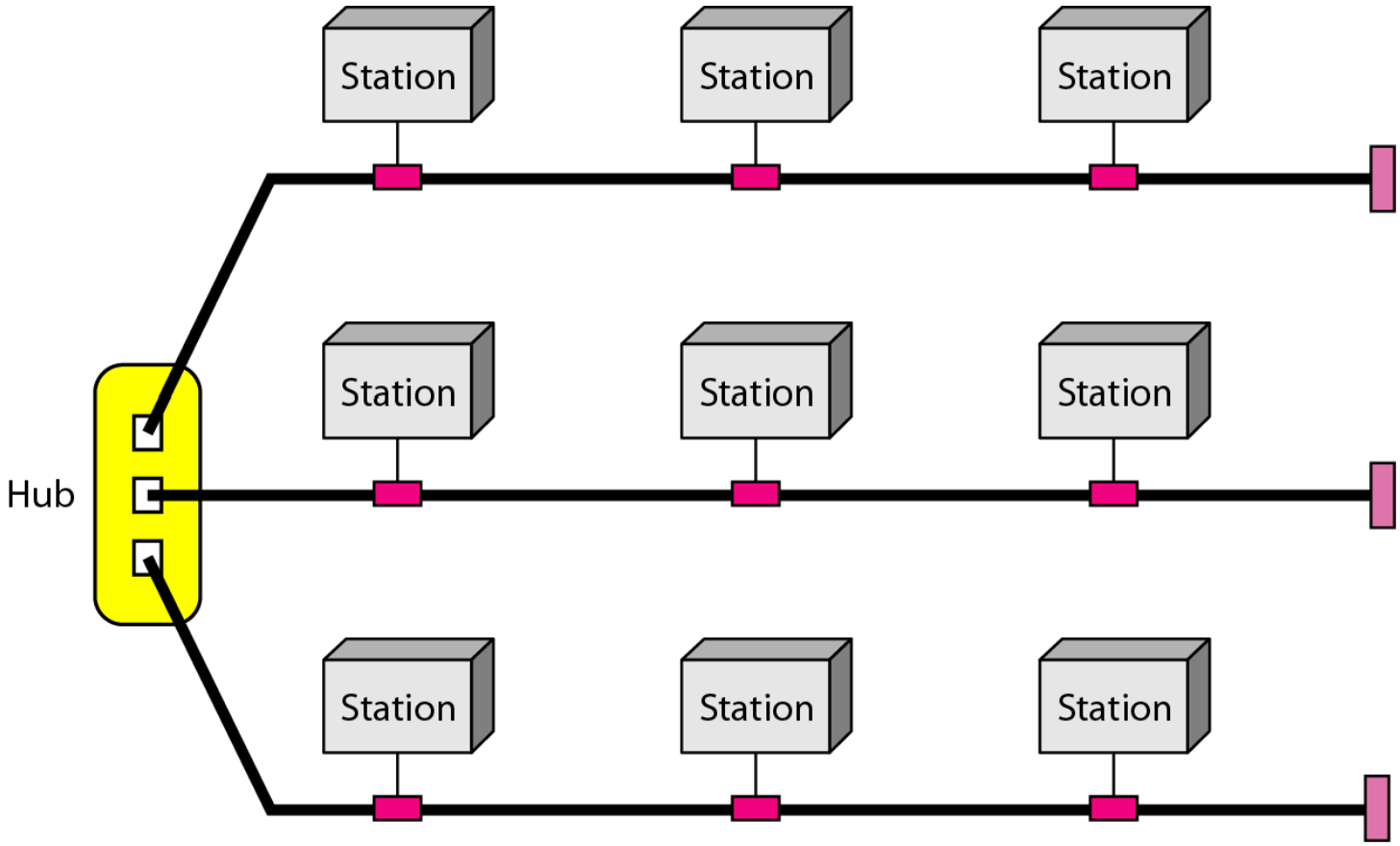
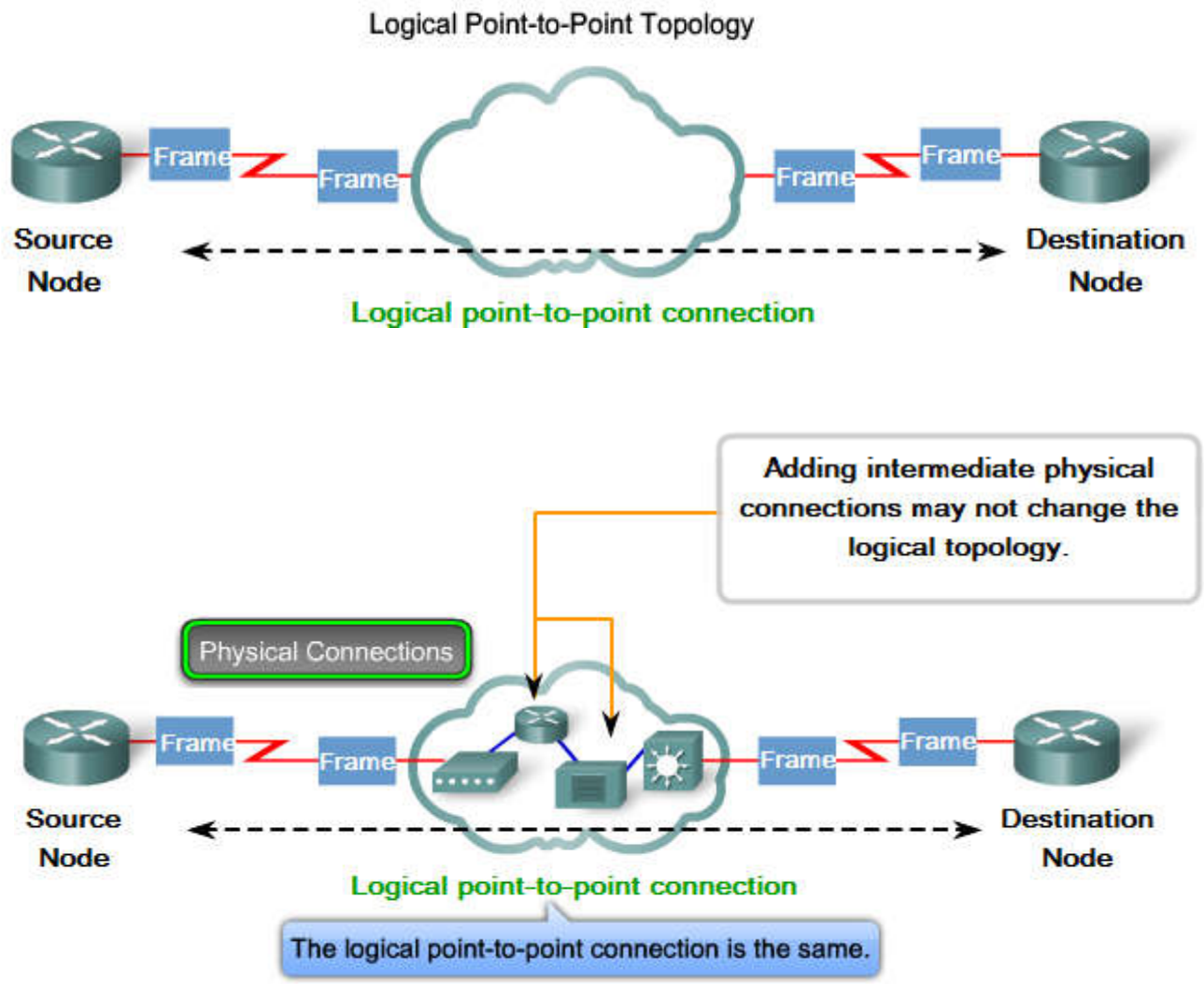


Figure 1.9 *A hybrid topology: a star backbone with three bus networks*

Logical Point-to-Point Topology

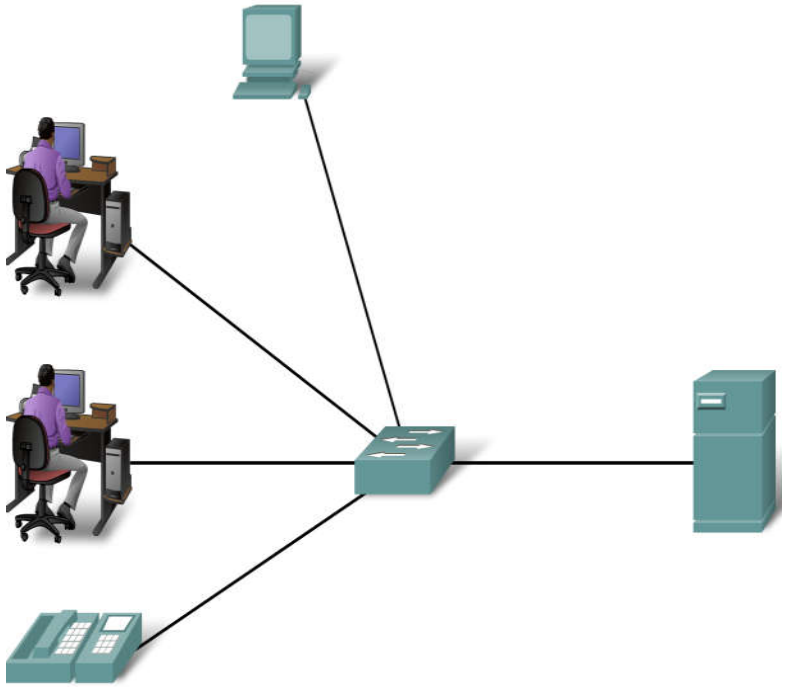
The end nodes communicating in a point-to-point network can be physically connected via a number of intermediate devices. However the use of physical devices in the network does not affect the logical topology. As shown in the figure, the source and destination node may be indirectly connected to each other over some geographical distance.



Categories of Network

Local Area Network (LAN):

A network serving a home, building or campus is considered a Local Area Network (LAN)



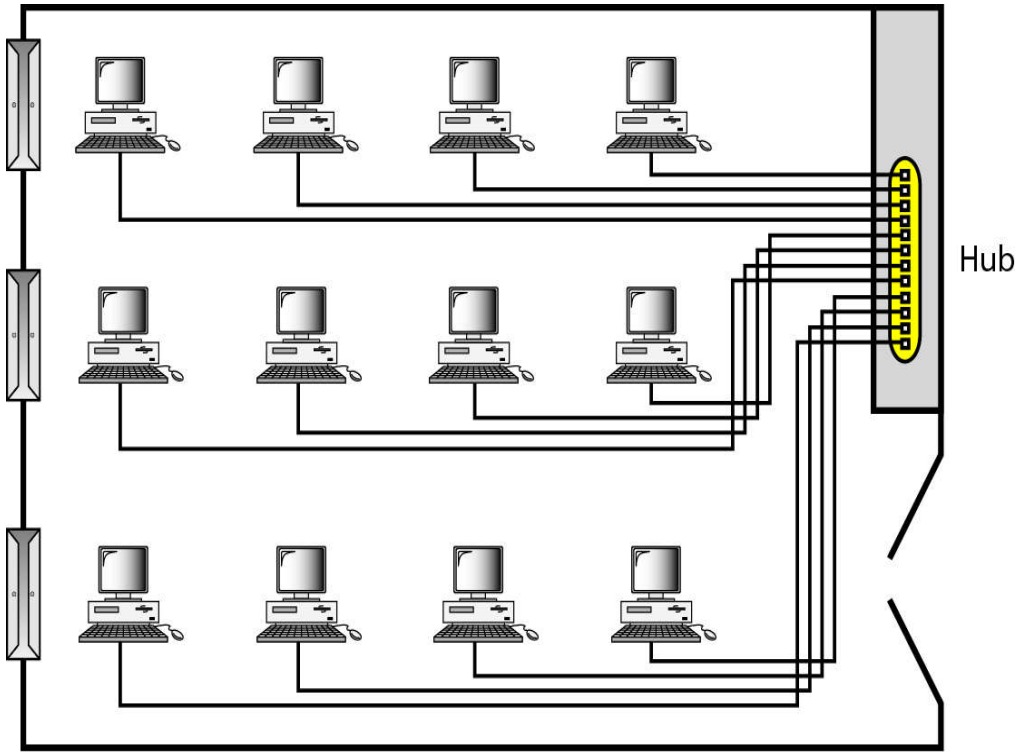
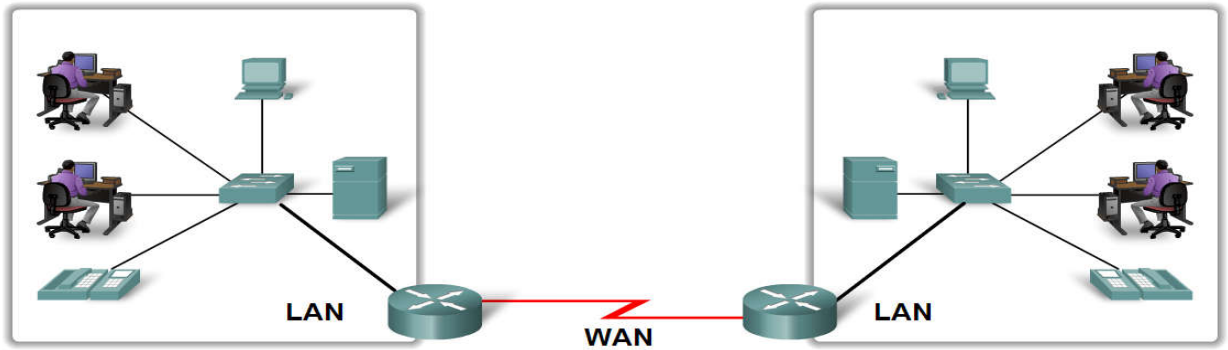


Figure 1.10 ***An isolated LAN connecting 12 computers to a hub in a closet***

Categories of Network

Wide Area Network (WAN):

LANs separated by geographic distance are connected by a network known as a Wide Area Network (WAN)



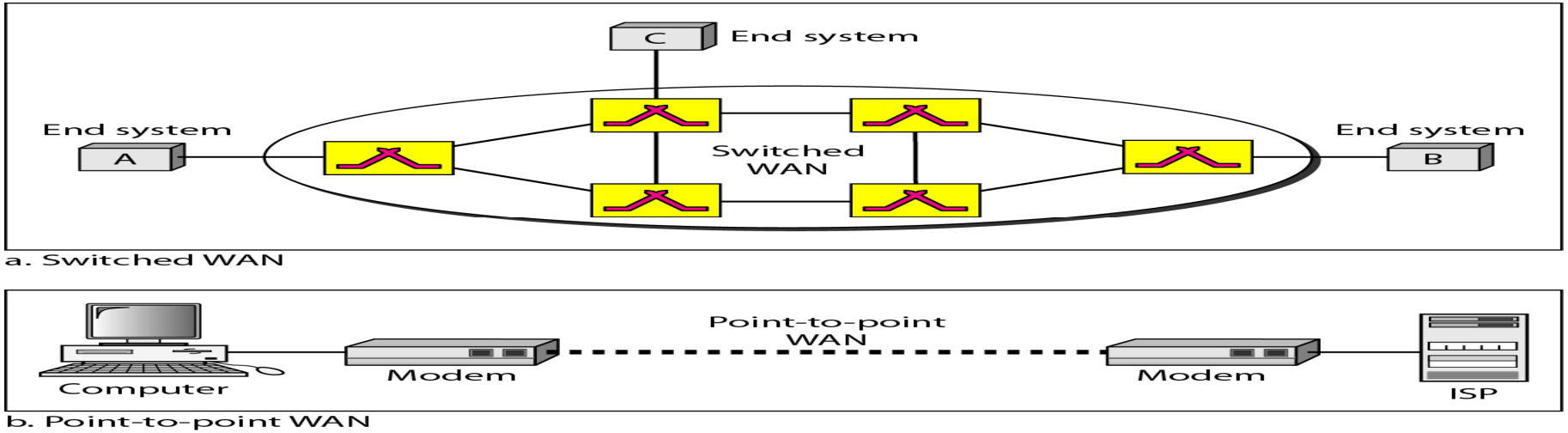


Figure 1.11 *WANs: a switched WAN and a point-to-point WAN*

Interconnection of Networks: Internetwork or internet

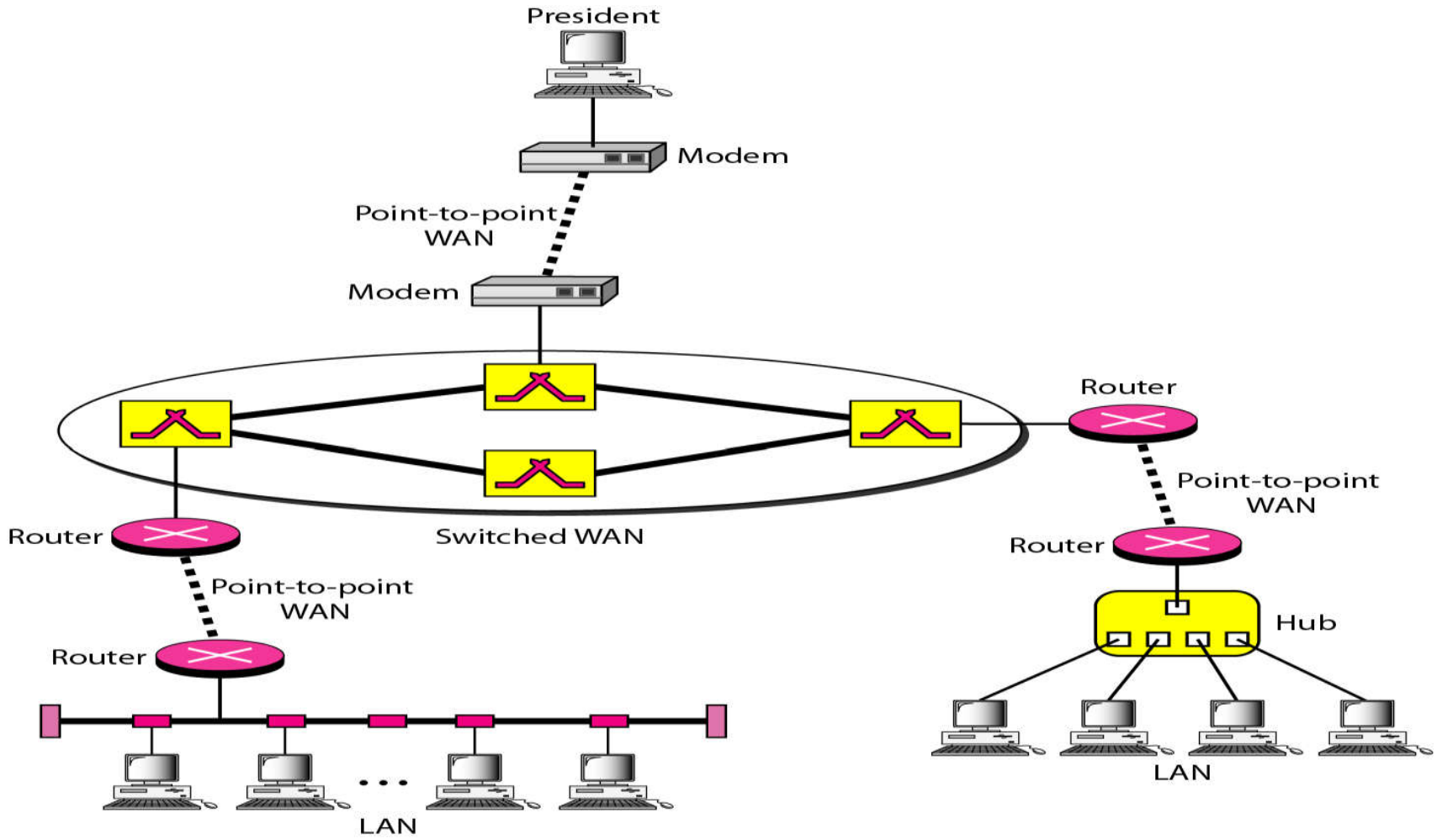
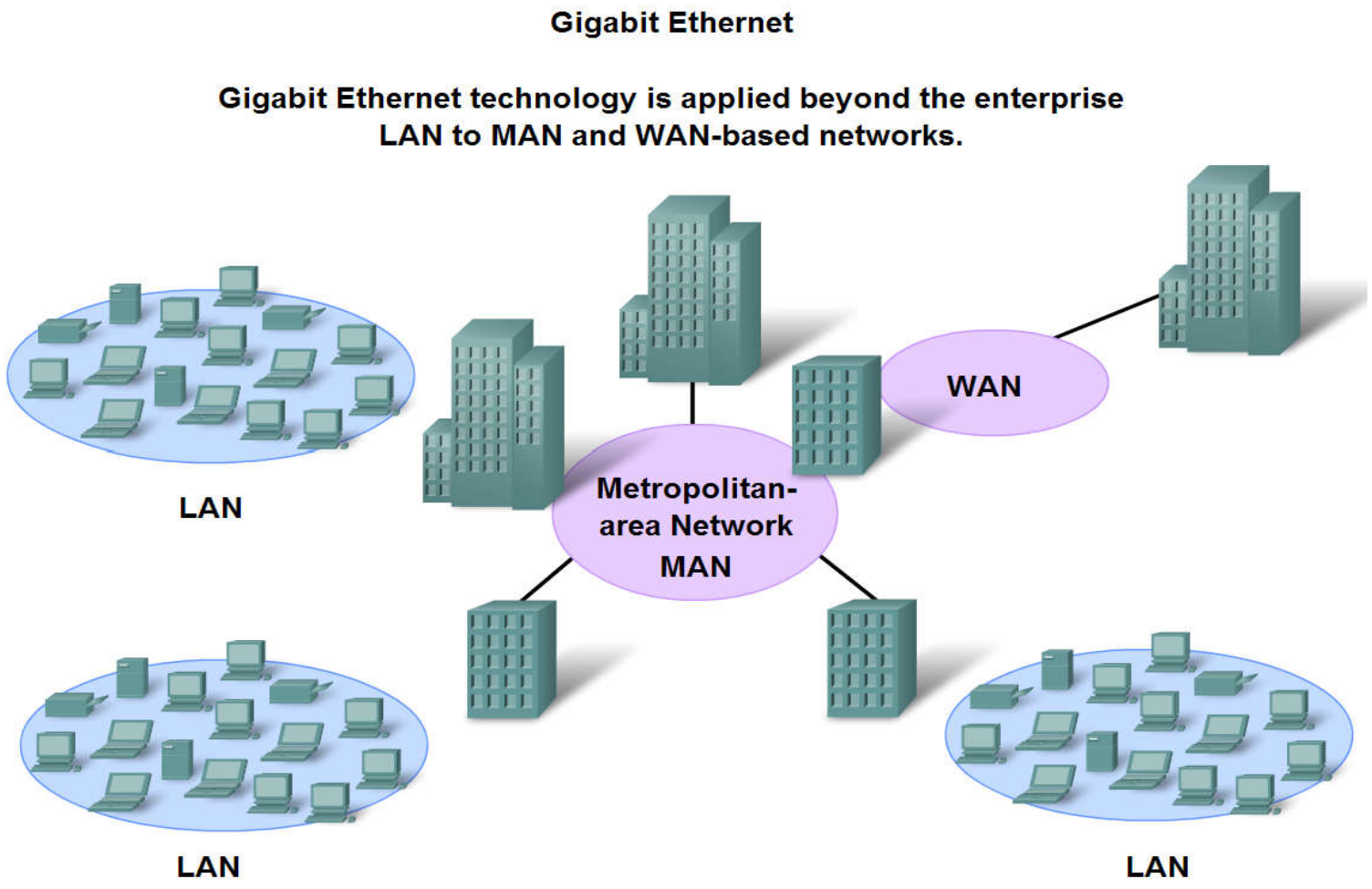


Figure 1.12 ***A heterogeneous network made of four WANs and two LANs***

Interconnection of Networks: Internetwork or internet



The Internet

The Internet has revolutionized many aspects of our daily lives. It has affected the way we do business as well as the way we spend our leisure time. The Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use.

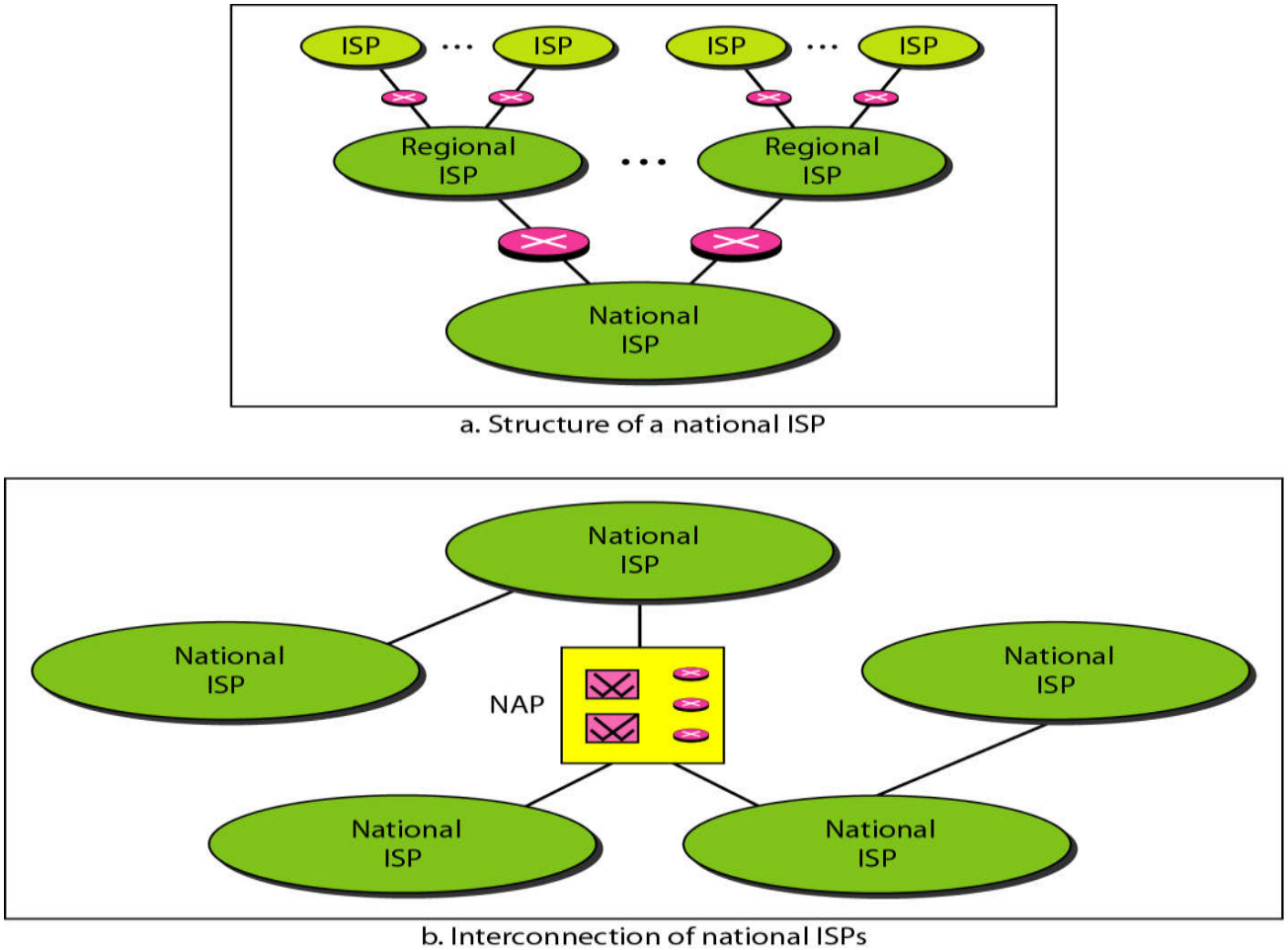
The Internet is the network of networks. It is a public network.

Topics discussed in this section:

The Internet Today (ISPs)

Reference: Behrouz A. Forouzan, *Data Communications and Networking*, Article 1.3

The Internet Today (ISPs)

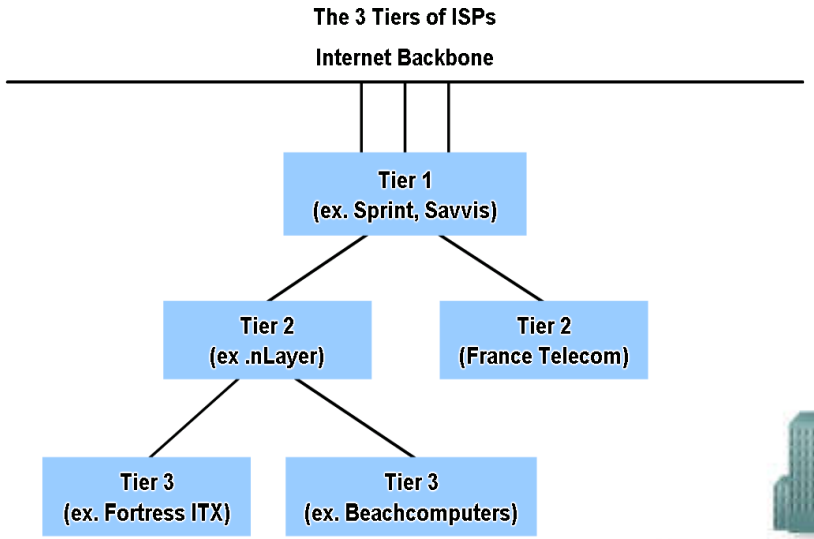


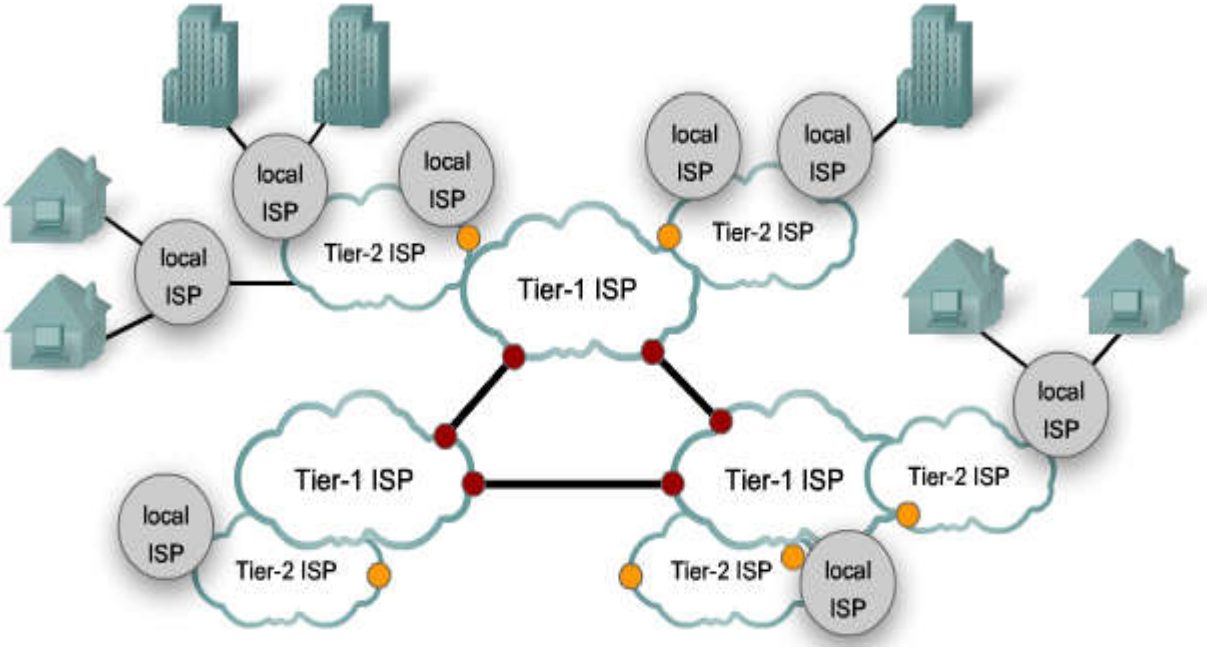
a. Structure of a national ISP

b. Interconnection of national ISPs

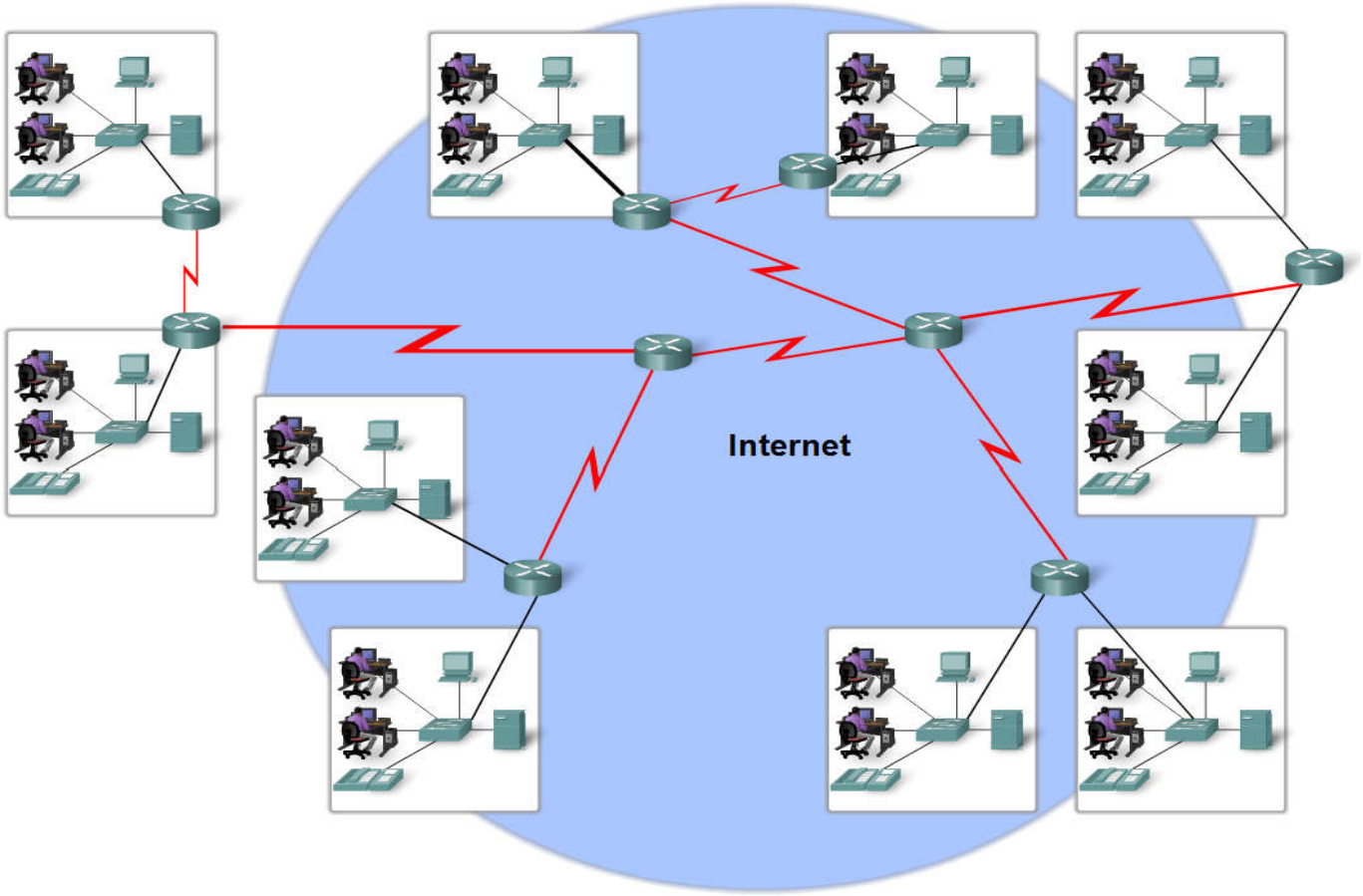
Figure 1.13 ***Hierarchical organization of the Internet***

The Internet Today (ISPs)

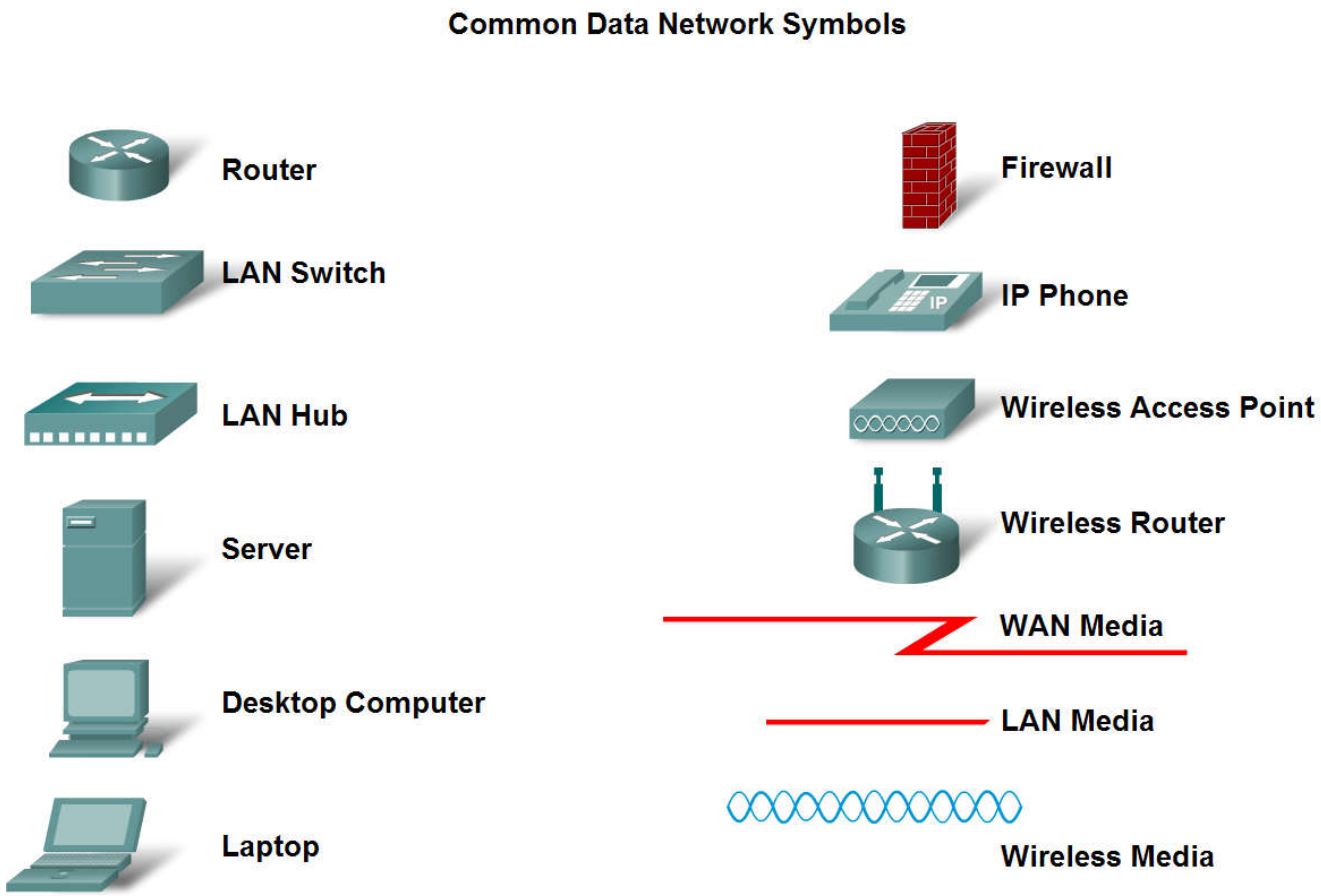




The Internet



Describe network representations



Protocols and Standards

Protocols:

- Protocol can be defined as a set of rules.
- Some protocols are TCP/IP, UDP, ICMP, FTP, NTP etc.

Standards:

- Standards are agreed-upon rules.
- Standards are essential in creating and maintaining an open and competitive market for equipment manufacturers and in guaranteeing national and international interoperability of data and telecommunications technology and processes.
- Standards provide guidelines to manufacturers, vendors, government agencies, and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications.
- Some standard creation committees are ISO, ANSI, IEEE etc.

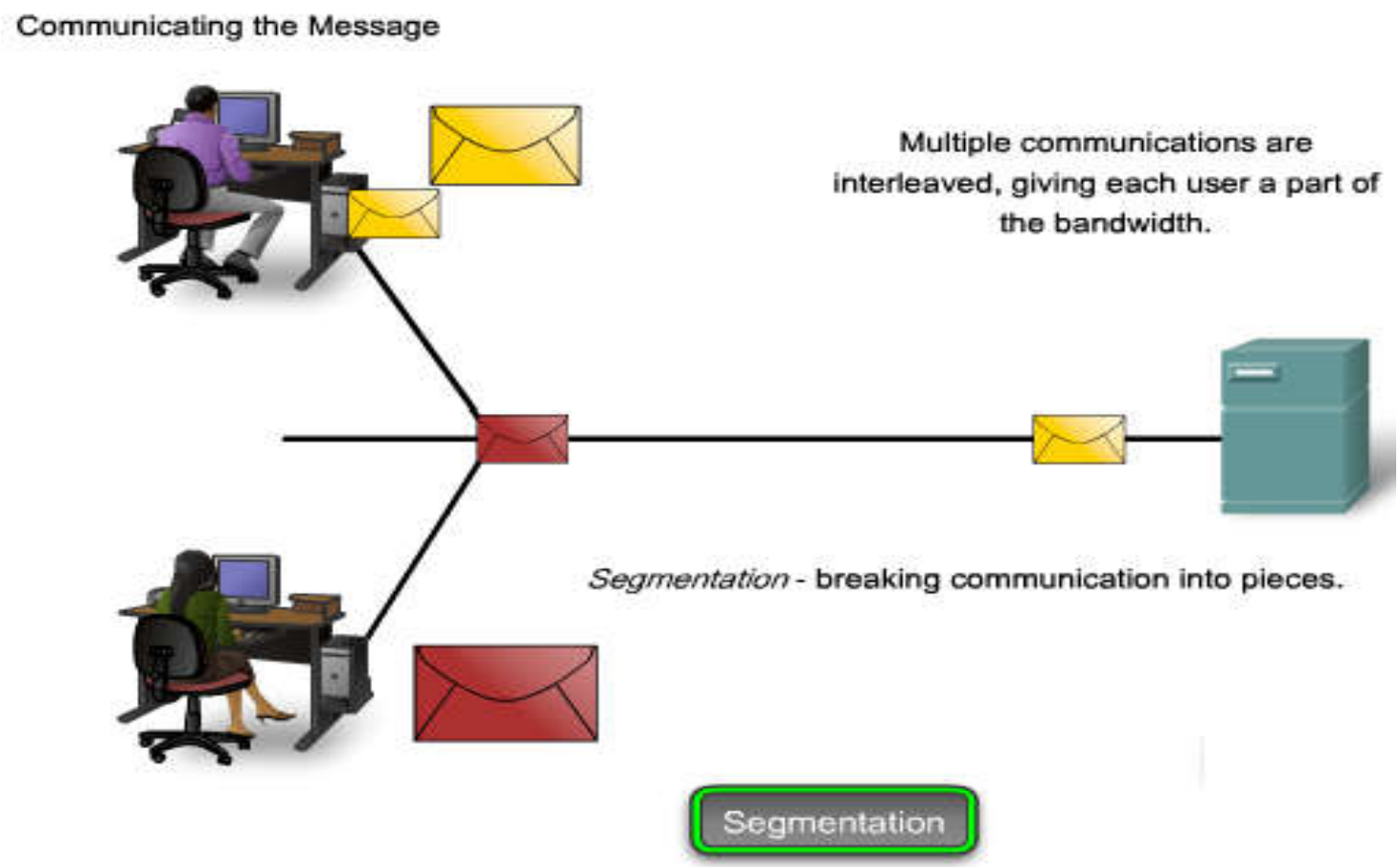
Reference: Behrouz A. Forouzan, *Data Communications and Networking*, Article 1.4

Network Basic

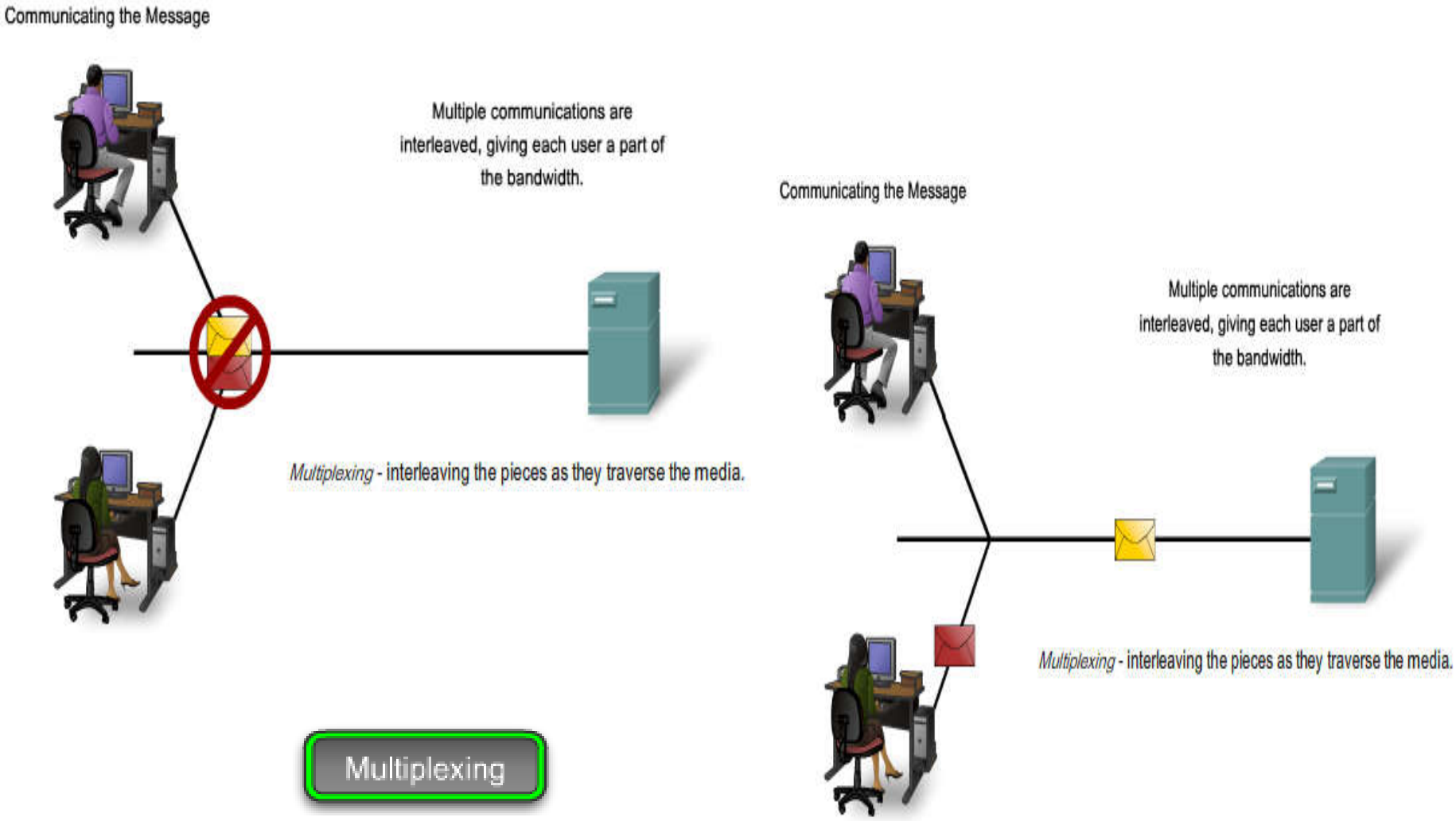
Communication the Message – Segmentation and Multiplexing
Quality of Communication
Converged Networks
The Network Architecture
- Fault Tolerance
- Scalability
- Quality of Service (QoS)
- Security

Circuit Switched Connection-Oriented Networks
Packet Switched Connectionless Networks
Communication between Sending and Receiving Device

Communicating the Message – Segmentation

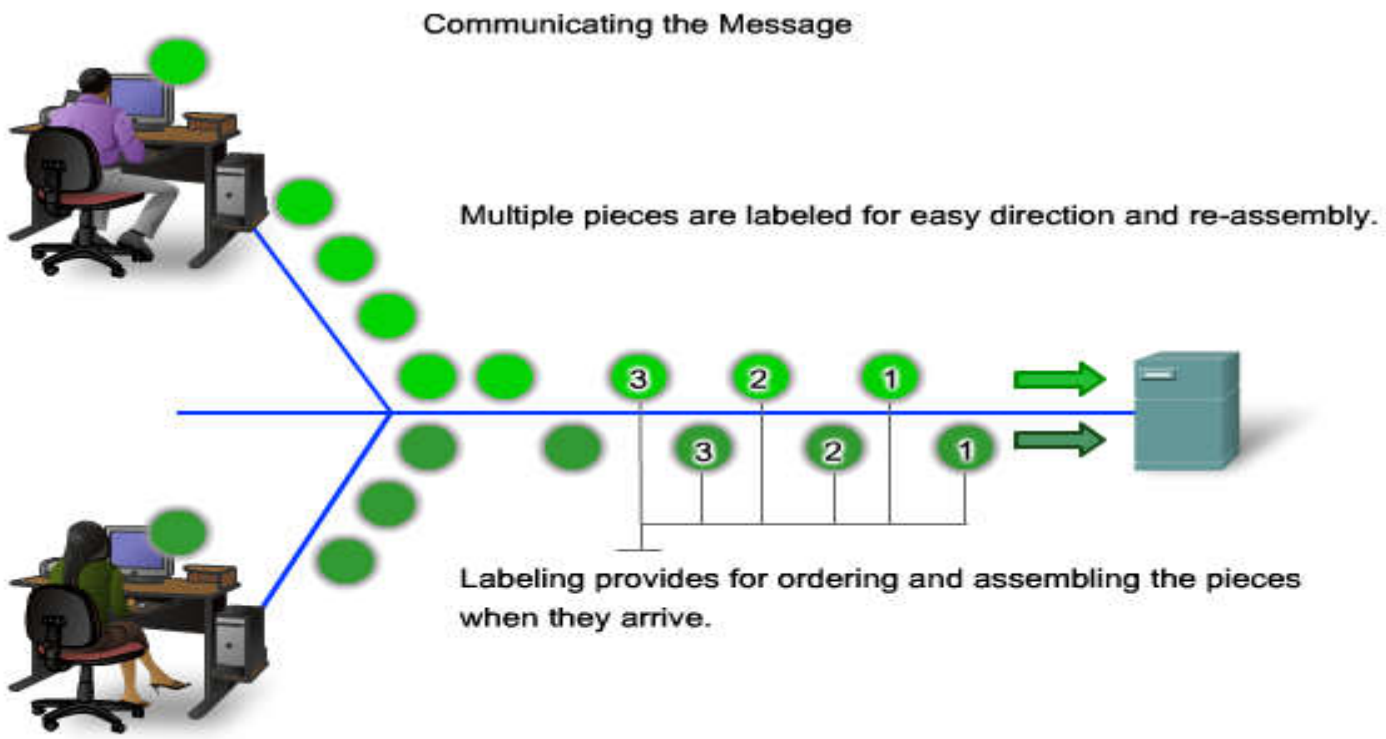


Communicating the Message - Multiplexing



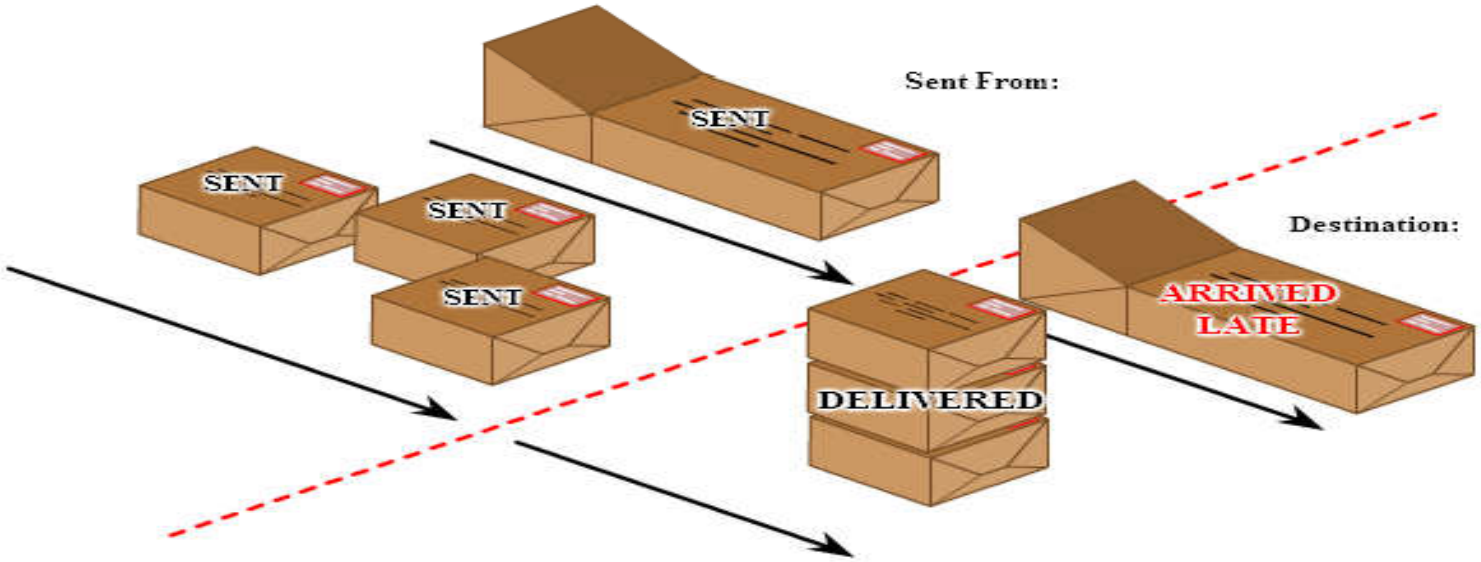
Communicating the Message - Multiplexing

- The downside to using segmentation and multiplexing to transmit messages across a network is the level of complexity that is added to the process. Imagine if you had to send a 100-page letter, but each envelope would only hold one page. The process of addressing, labeling, sending, receiving, and opening the entire hundred envelopes would be time-consuming for both the sender and the recipient.
- In network communications, each segment of the message must go through a similar process to ensure that it gets to the correct destination and can be reassembled into the content of the original message.



Quality of Communication

- Different types of messages may vary in complexity and importance. Clear and concise messages are usually easier to understand than complex messages. Important communications require more care to ensure that they are delivered and understood by the recipient.
- The factors that affecting the successful communication across the network include:
- The size of the message
- The complexity of the message
- The importance of the message
- Large messages may be interrupted or delayed at different points within the network. A message with a low importance or priority could be dropped if the network becomes overloaded.



It is more difficult to deliver a large bulky package, quickly and without damage, than it is to deliver a number of smaller, less complex packages.

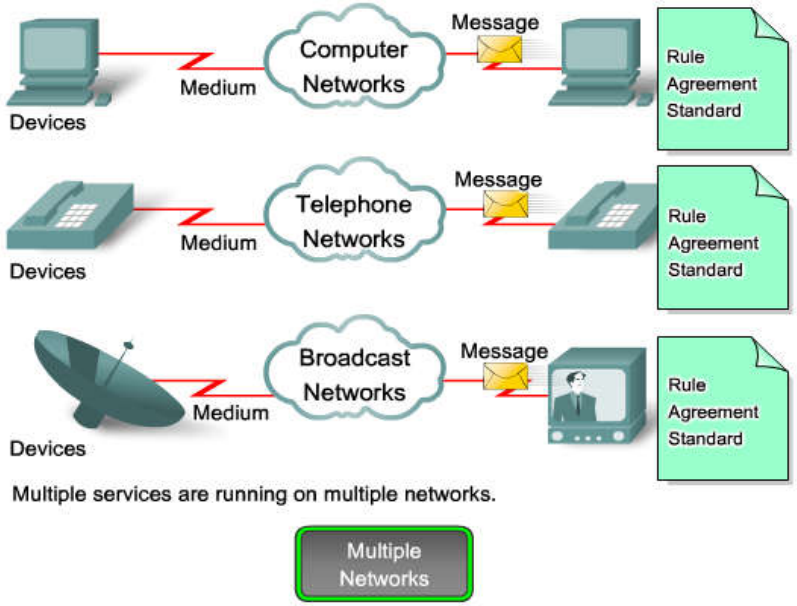
Converged Networks

Multiple services-multiple networks:

Traditional telephone, radio, television, and computer data networks each have their own individual versions of the four basic network elements. In the past, every one of these services required a different technology to carry its particular communication signal. Additionally, each service had its own set of rules and standards to ensure successful communication of its signal across a specific medium.

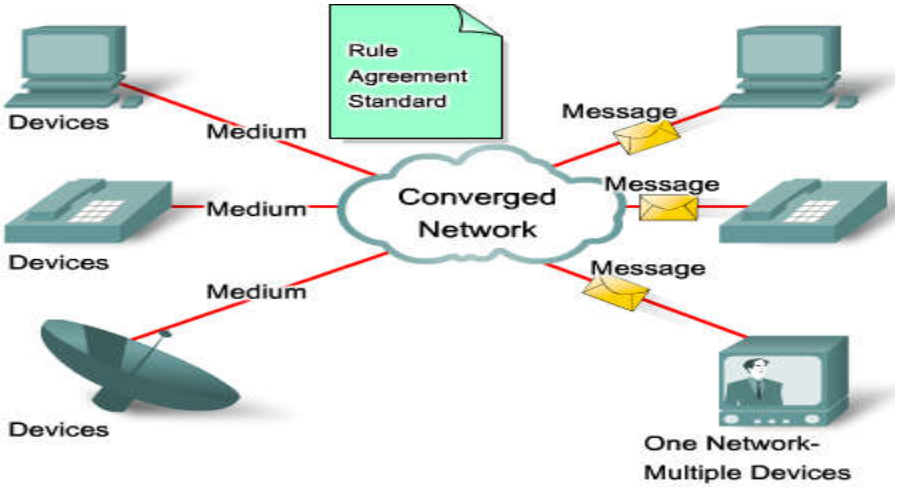
Converged networks:

Technology advances are enabling us to consolidate these disparate networks onto one platform - a platform defined as a converged network. The flow of voice, video, and data traveling over the same network eliminates the need to create and maintain separate networks. On a converged network there are still many points of contact and many specialized devices - for example, personal computers, phones, TVs, personal assistants, and retail point-of-sale registers - but only one common network infrastructure.



Multiple services are running on multiple networks.

Multiple Networks

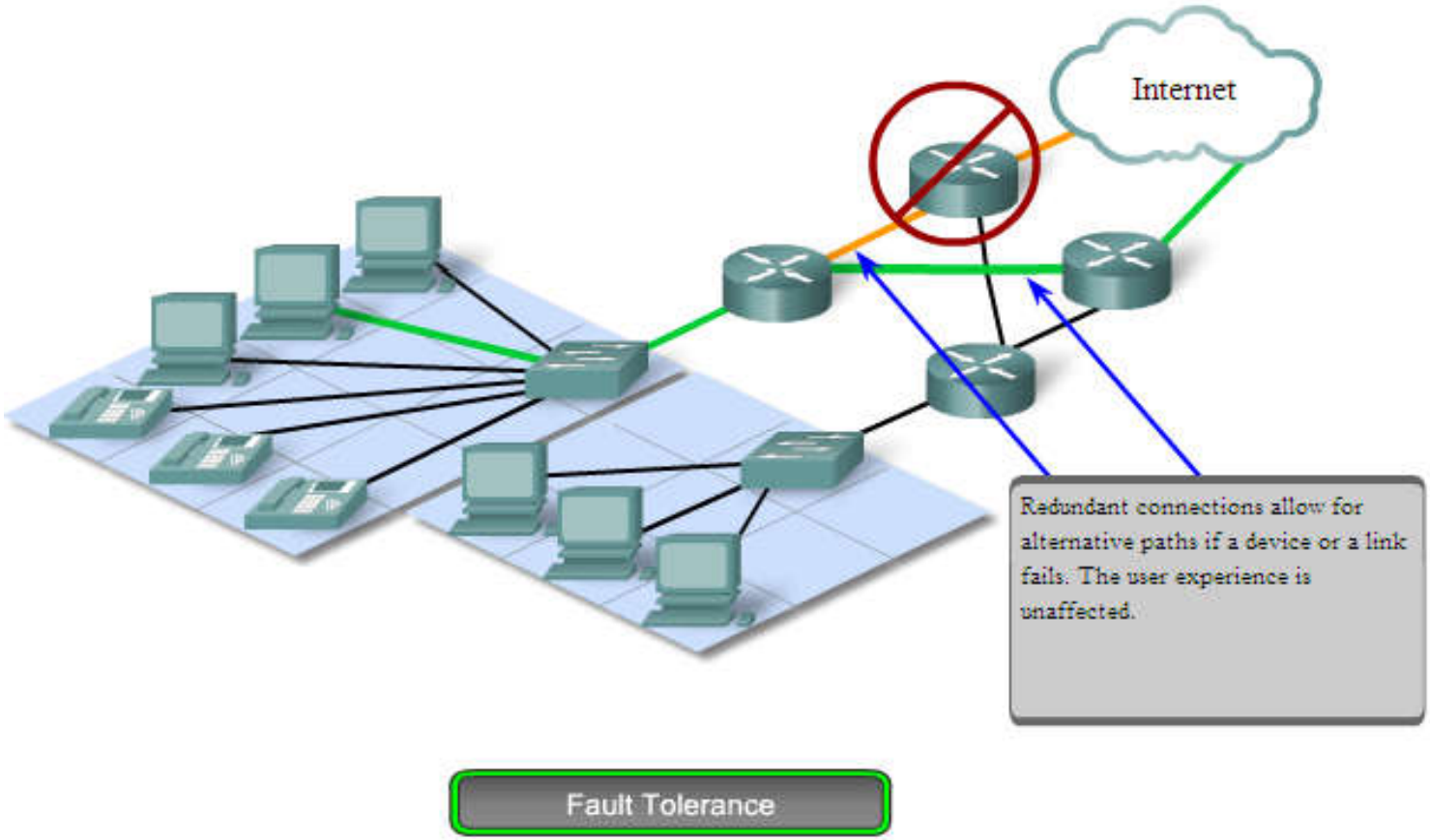


Converged data networks carry multiple services on one network.

Converged Networks

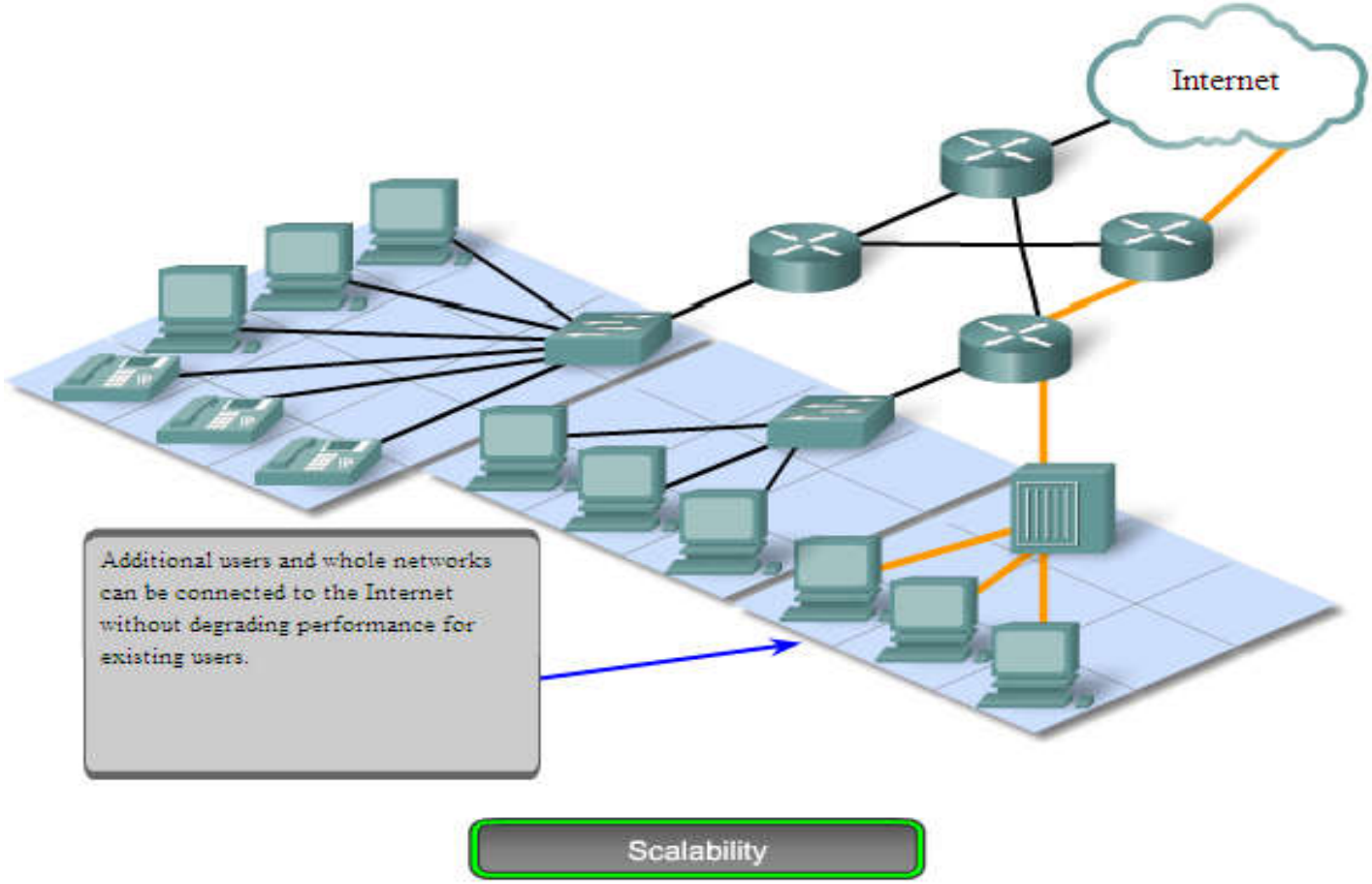
The Network Architecture – Fault Tolerance

A fault tolerant network is one that limits the impact of a hardware or software failure and can recover quickly when such a failure occurs. These networks depend on redundant links, or paths, between the source and destination of a message. If one link or path fails, processes ensure that messages can be instantly routed over a different link transparent to the users on either end. Both the physical infrastructures and the logical processes that direct the messages through the network are designed to accommodate this redundancy.



The Network Architecture – Scalability

A scalable network can expand quickly to support new users and applications without impacting the performance of the service being delivered to existing users. Thousands of new users and service providers connect to the Internet each week. The ability of the network to support these new interconnections depends on a hierarchical layered design for the underlying physical infrastructure and logical architecture.

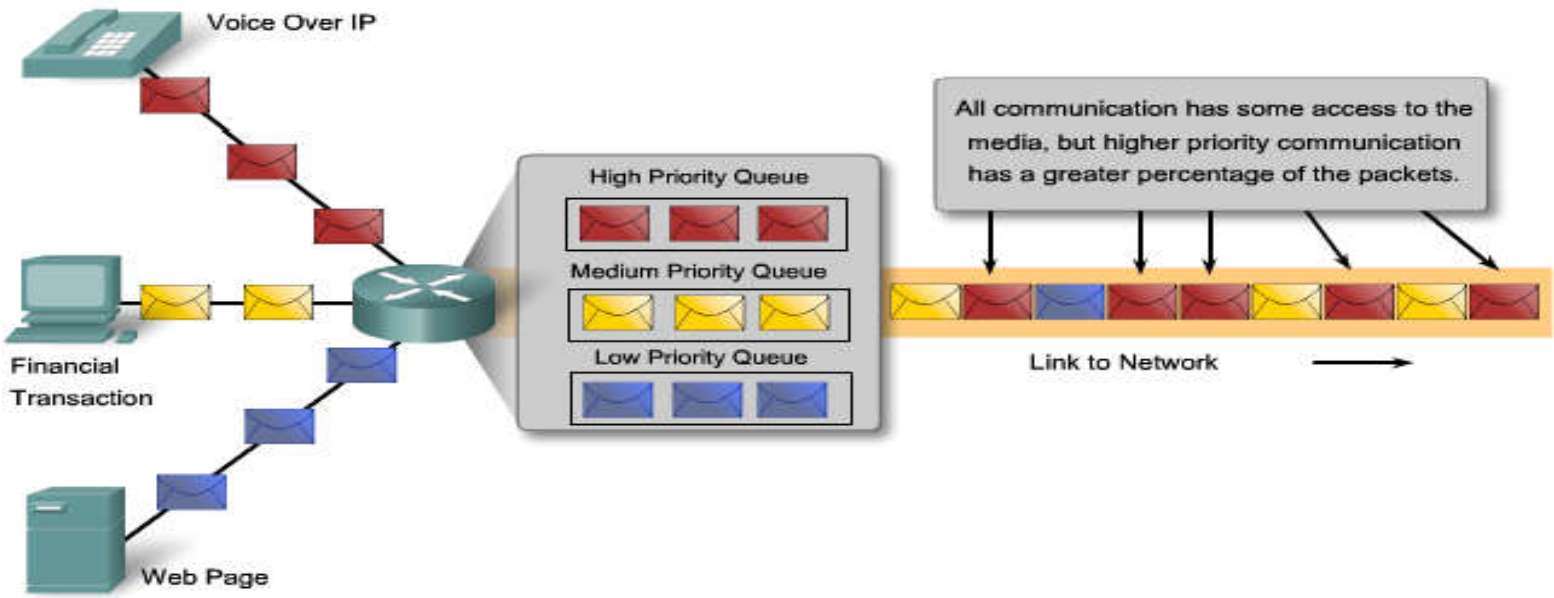


The Network Architecture – Quality of Service (QoS)

Providing Quality of Service:

Achieving the required Quality of Service (QoS) by managing the delay and packet loss parameters on a network becomes the secret to a successful end-to-end application quality solution. Thus, ensuring QoS requires a set of techniques to manage the utilization of network resources. In order to maintain a high quality of service for applications that require it, it is necessary to prioritize which types of data packets must be delivered at the expense of other types of packets that can be delayed or dropped.

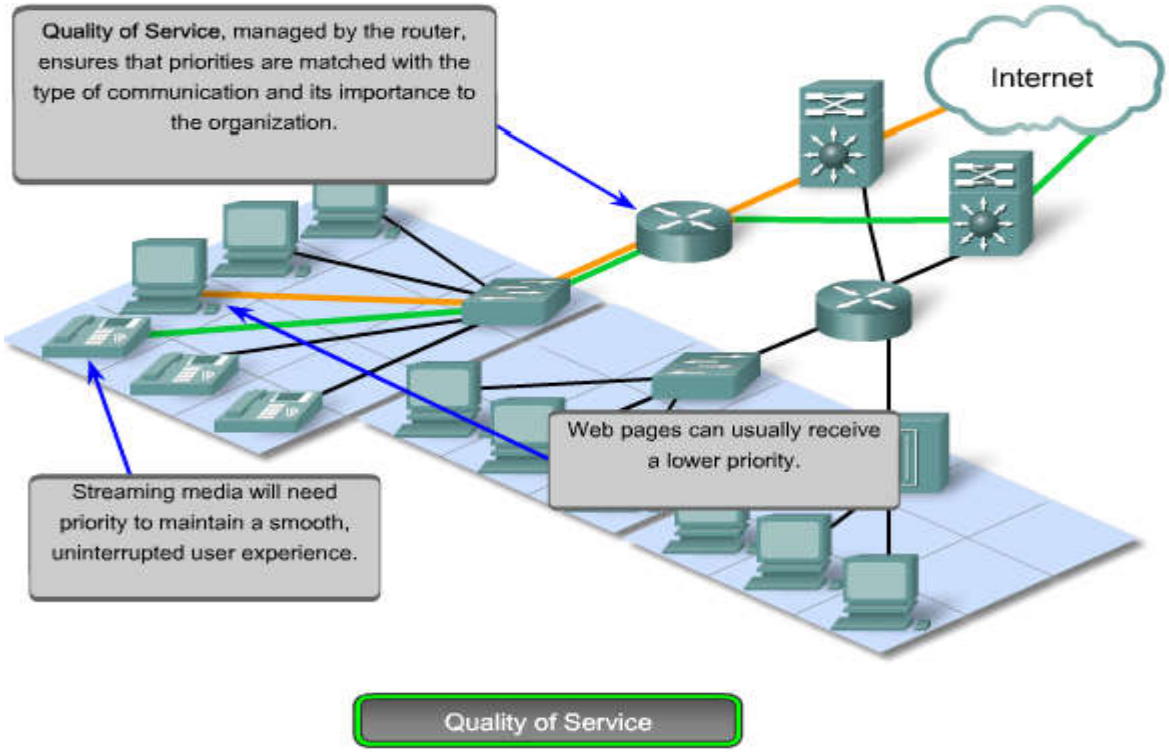
Using Queues to Prioritize Communication



Queuing according to data type enables voice data to have priority over transaction data, which has priority over web data.

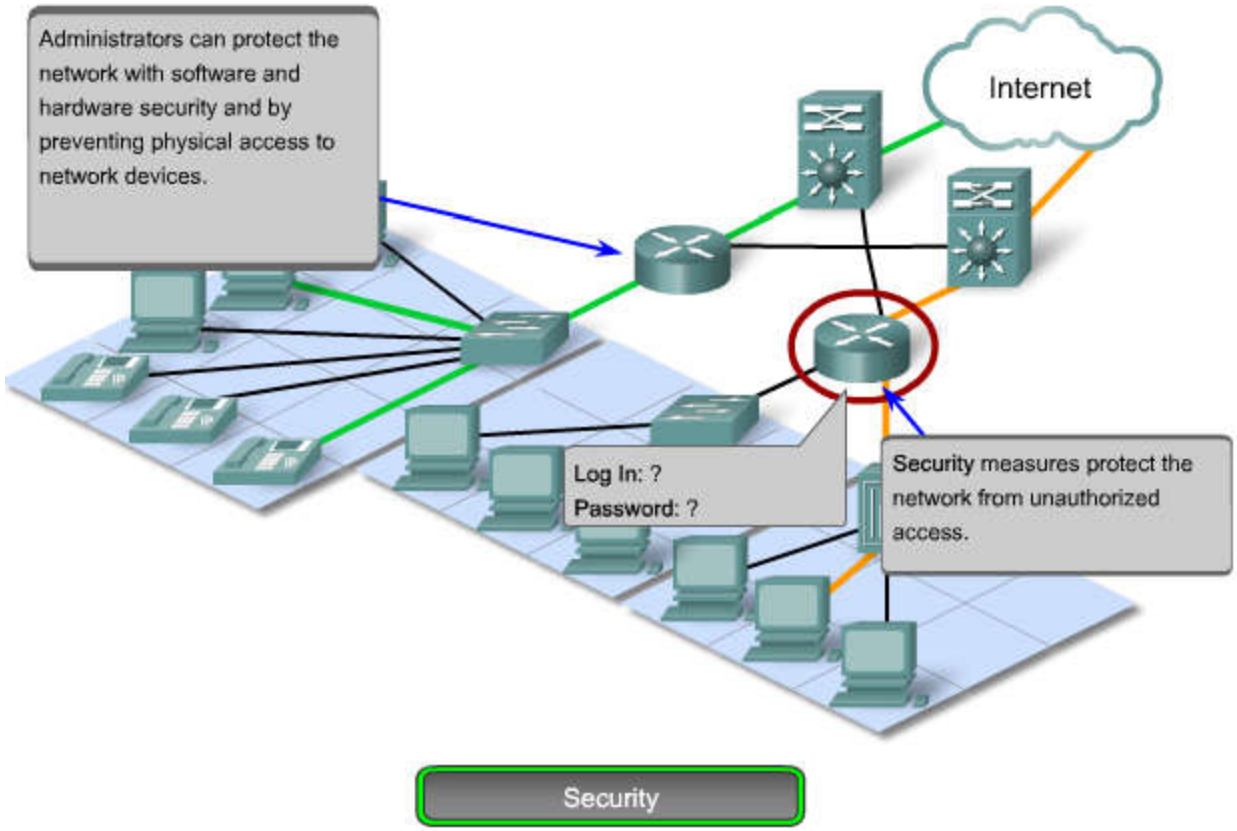
The Network Architecture – Quality of Service (QoS)

The Internet is currently providing an acceptable level of fault tolerance and scalability for its users. But new applications available to users over internetworks create higher expectations for the quality of the delivered services. Voice and live video transmissions require a level of consistent quality and uninterrupted delivery that was not necessary for traditional computer applications. Quality of these services is measured against the quality of experiencing the same audio or video presentation in person. Traditional voice and video networks are designed to support a single type of transmission, and are therefore able to produce an acceptable level of quality. New requirements to support this quality of service over a converged network are changing the way network architectures are designed and implemented.



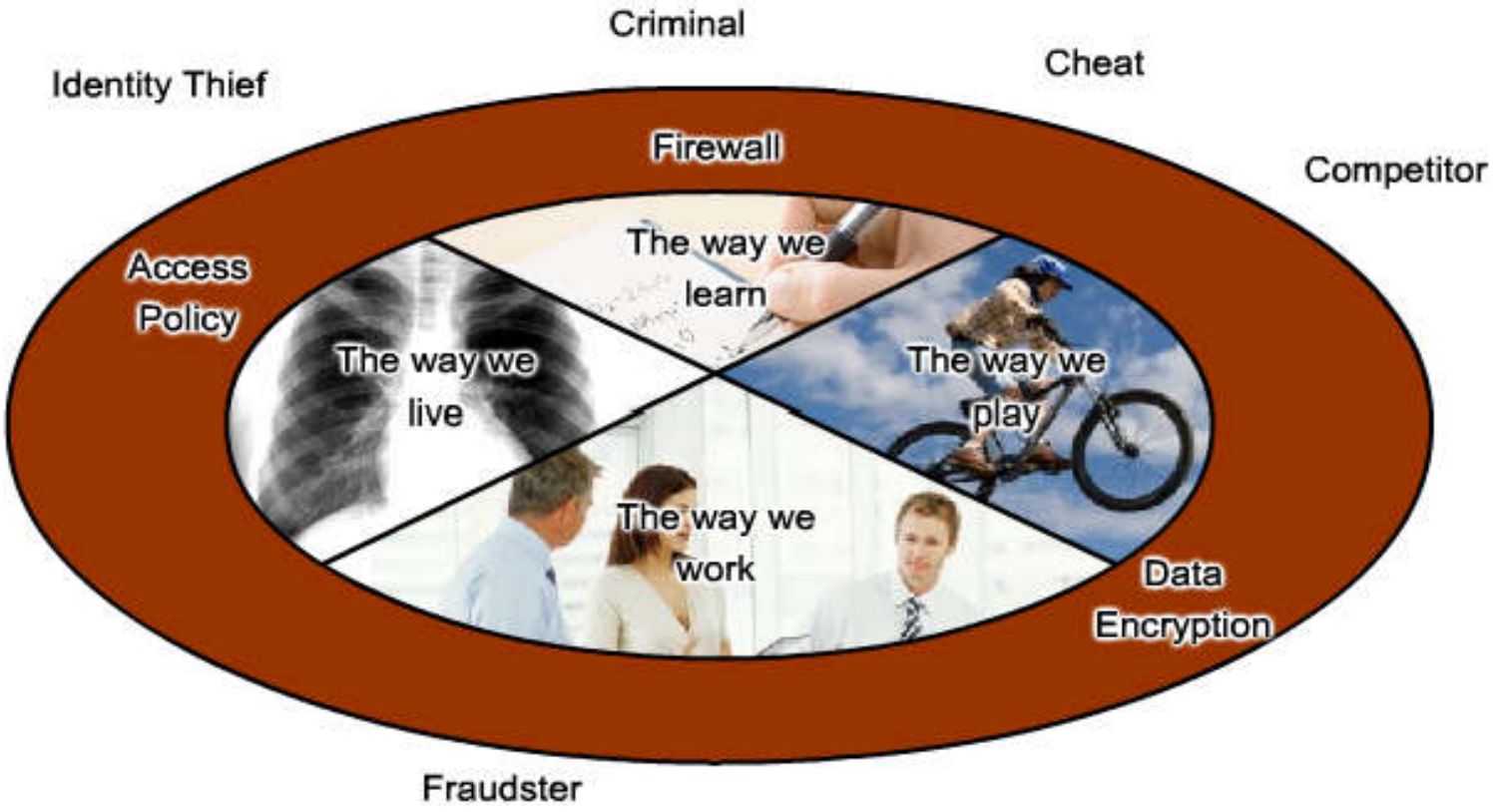
The Network Architecture – Security

The Internet has evolved from a tightly controlled internetwork of educational and government organizations to a widely accessible means for transmission of business and personal communications. As a result, the security requirements of the network have changed. The security and privacy expectations that result from the use of internetworks to exchange confidential and business critical information exceed what the current architecture can deliver. Rapid expansion in communication areas that were not served by traditional data networks is increasing the need to embed security into the network architecture



The Network Architecture – Security

Providing Network Security:

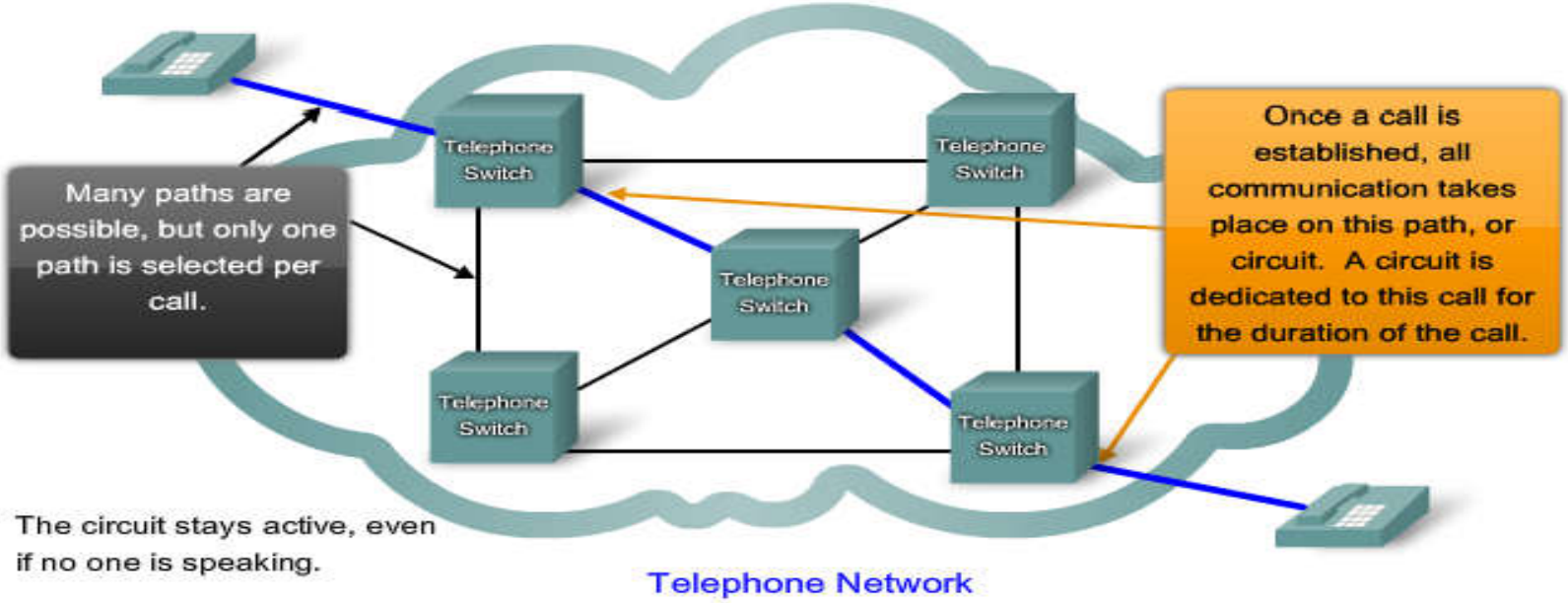


The communication and information that we would like to be private is protected from those who would make unauthorized use of it.

Circuit Switched Connection-Oriented Networks

To understand this it is necessary to look at how early telephone systems work. When a person makes a call using a traditional telephone set, the call first goes through a setup process, where all of the telephone switching locations between the person and the phone set that they are calling are identified. A temporary path, or circuit, is created through the various switching locations to use for the duration of the telephone call. If any link or device participating in the circuit fails, the call is dropped. To reconnect, a new call must be made, and a new circuit created between the source telephone set and the destination. This type of connection-oriented network is called a circuit-switched network.

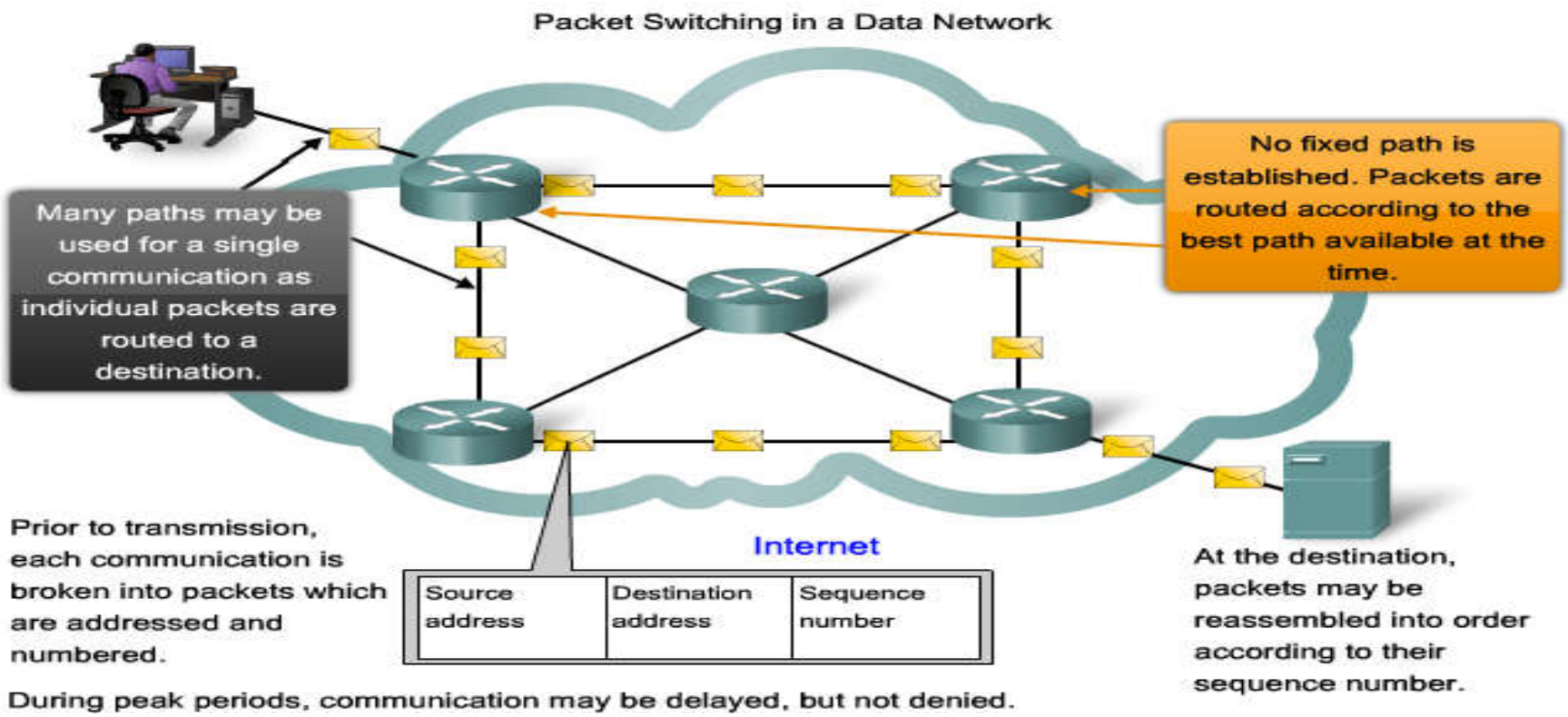
Circuit Switching in a Telephone Network



There are many, many circuits, but a finite number. During peak periods, some calls may be denied.

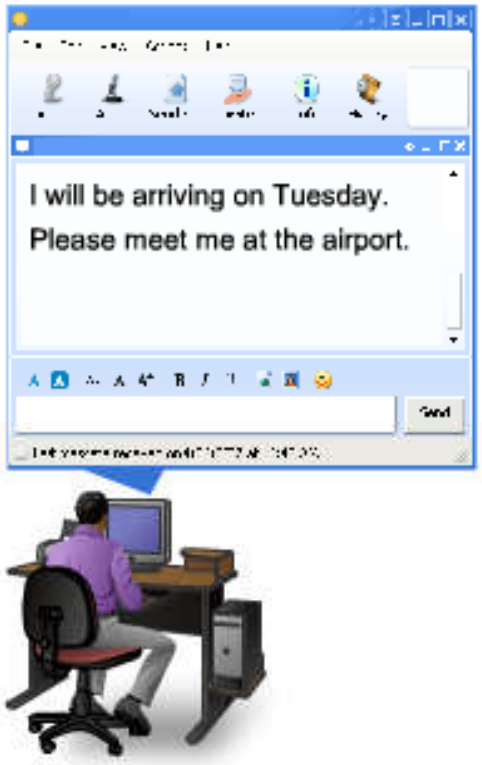
Packet Switched Connectionless Networks

In the search for a network that could withstand the loss of a significant amount of its transmission and switching facilities, the early Internet designers reevaluated early research regarding packet switched networks. The premise for this type of networks is that a single message can be broken into multiple message blocks. Individual blocks containing addressing information indicate both their origination point and their final destination. Using this embedded information, these message blocks, called packets, can be sent through the network along various paths, and can be reassembled into the original message upon reaching their destination.



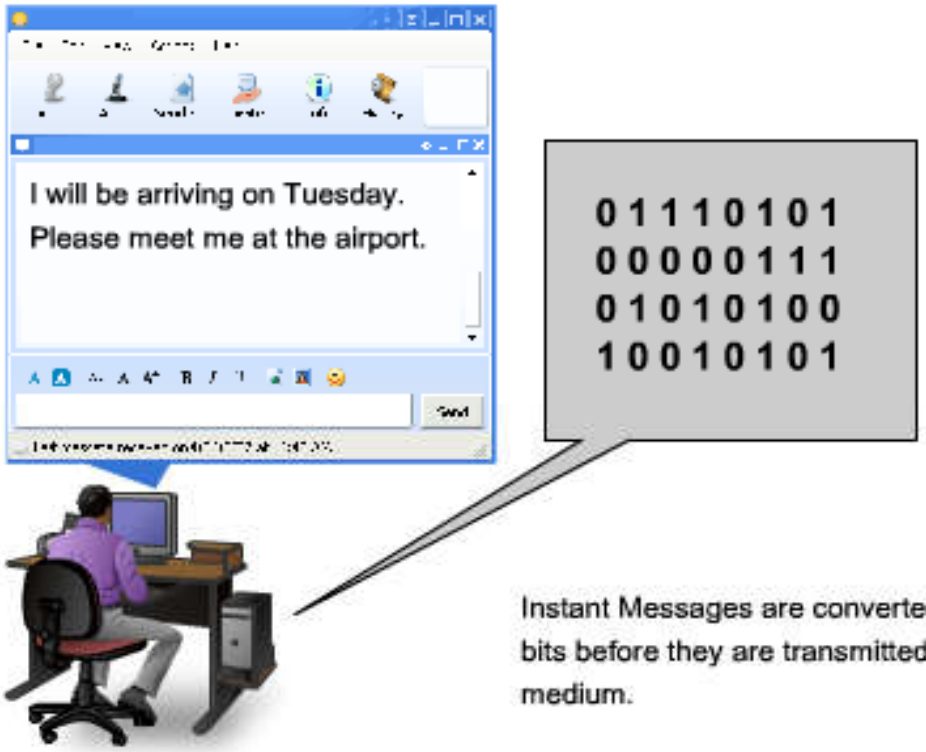
Communication between Sending and Receiving Device

Sending an Instant Message



Communication between Sending and Receiving Device

Sending an Instant Message

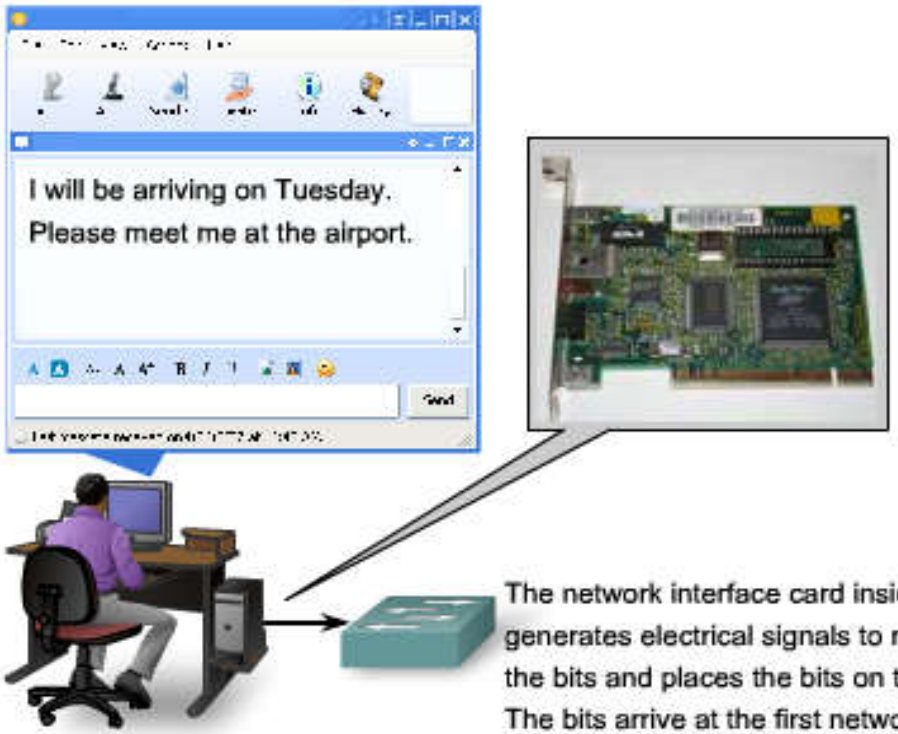


Instant Messages are converted to binary bits before they are transmitted on the medium.

1 2 3 4 5 6 7

Communication between Sending and Receiving Device

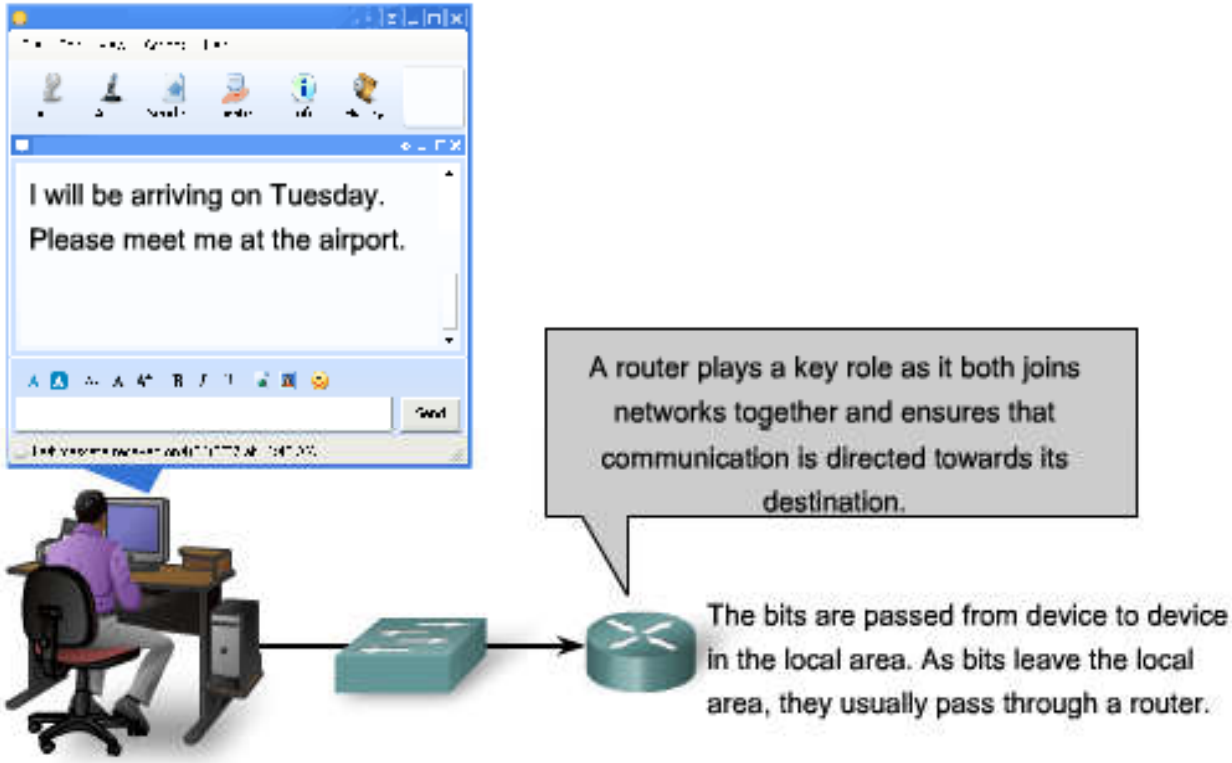
Sending an Instant Message



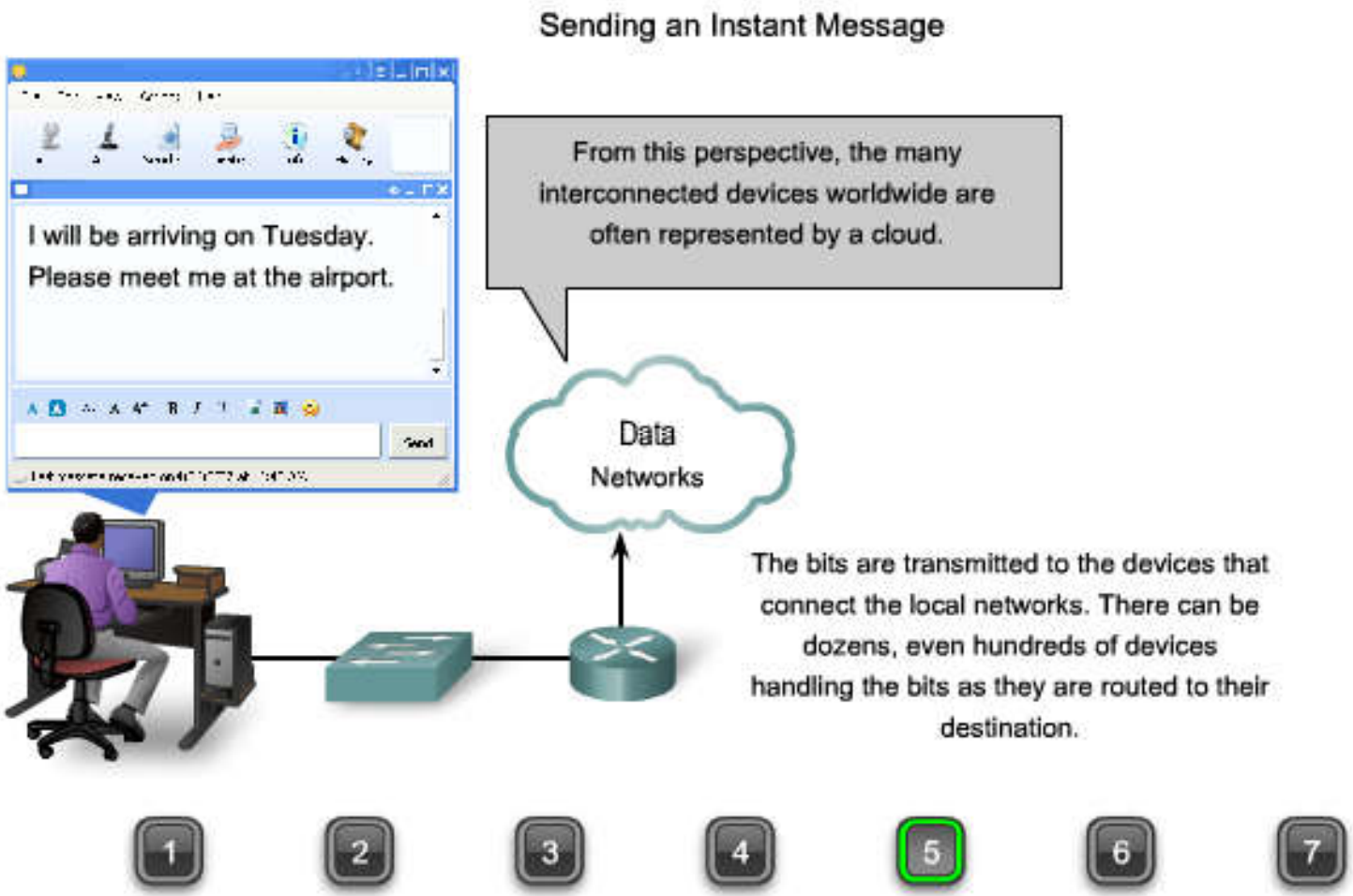
The network interface card inside the PC generates electrical signals to represent the bits and places the bits on the medium. The bits arrive at the first network device.

Communication between Sending and Receiving Device

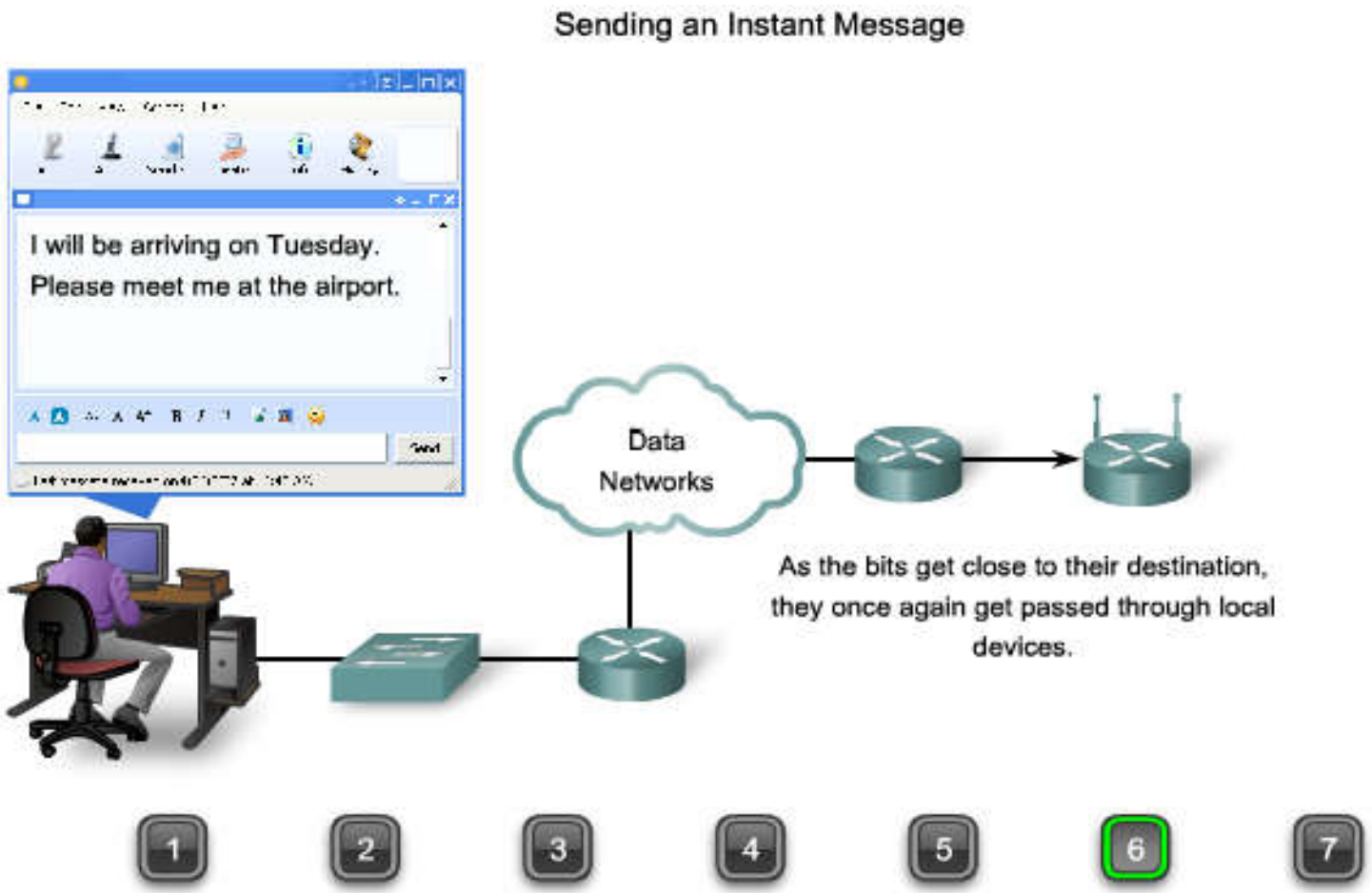
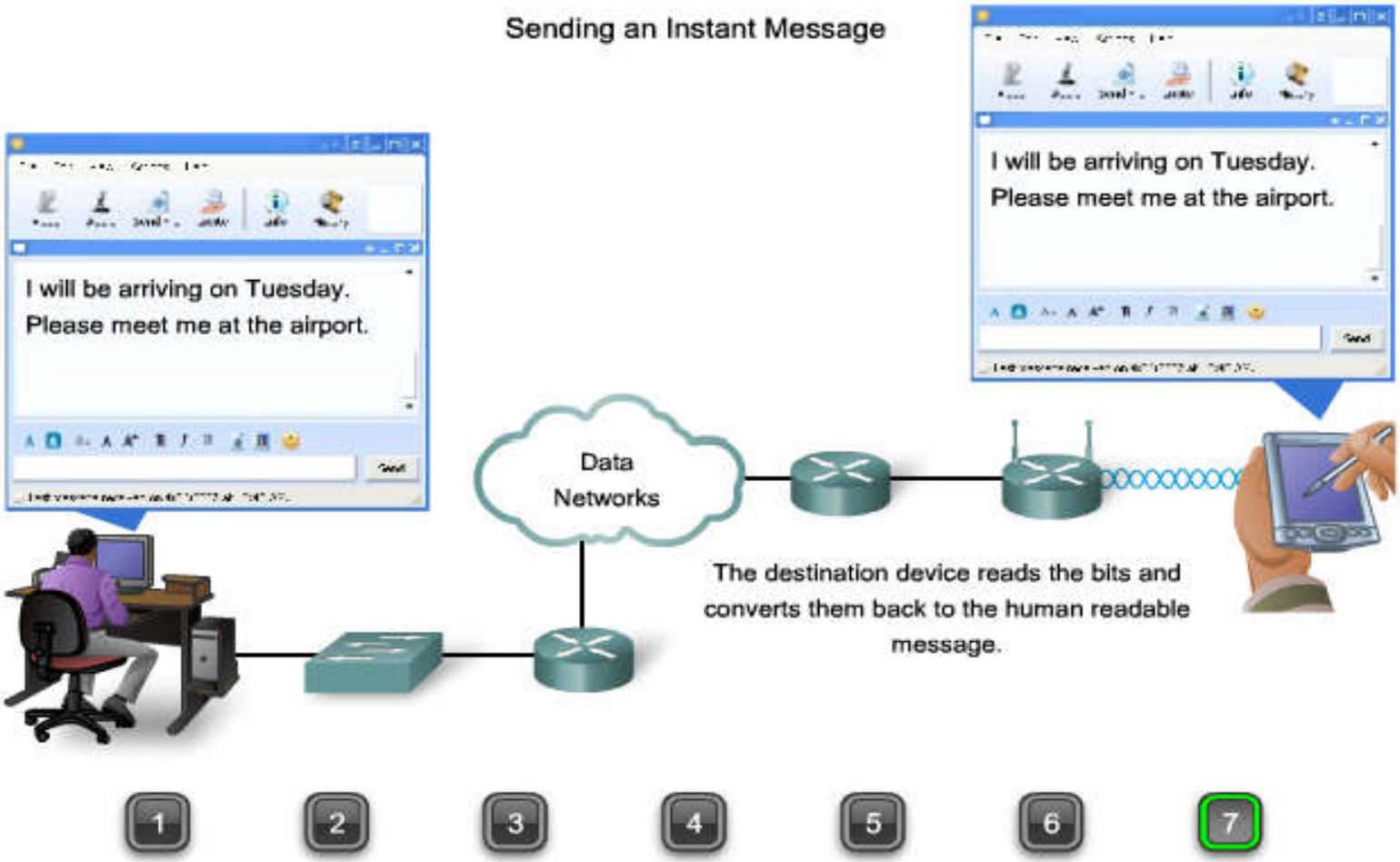
Sending an Instant Message



Communication between Sending and Receiving Device



Communication between Sending and Receiving Device



Communication between Sending and Receiving Device